OCEAN NUMERICAL LABORATORY

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NB: These notes are a working draft. They are preliminary, non-definitive, and incomplete. Most notes/discussions are from other Authors. The work so far has been in selecting different Authors for different arguments and trying to make them consistent with each other. There may be errors in the words, in the equations … everywhere! Students should refer to the following books for a proper/complete study:

- Haidvogel and Beckmann, Numerical Ocean Circulation Modeling, Imperial College Press, 1999.
- Benoit Cushman-Roisin and Jean-Marie Becker. Introduction to Geophysical Fluid Dynamics. Physical and Numerical Aspects. Academic Press.
- F. Mesinger, A. Arakawa, Numerical Methods Used in Atmospheric Models, GARP Publ. Ser. No. 17, vol. 1, WMO, Geneva, 1976

1. THE EQUATIONS OF MOTION

Fluids are governed by the ordinary laws of mechanics: the **conservation of mass** and the **conservation of momentum**. For a full description of the fluid, we must also consider the equation of state, relating the density of the fluid to thermodynamic variables such as pressure, temperature, and salinity. The equations describing the fluid arise from applying Newton's second law to fluid motion. After summing all the forces involved, the result equations are parabolic equations with useful analytic characteristics, the Navier-Stokes equations a set of partial differential equations which describe the motion of viscous fluid substances, named after Claude-Louis Navier and George Gabriel Stokes. The Navier–Stokes equations (NS equations here after) express **conservation of momentum** and **conservation of mass** for a fluid in which the viscous stresses arising from its flow are linearly correlated to the local strain rate — a Newtonian fluid. The solution of the equations is a flow velocity in the Eulerian space; that is the solution is given for fixed points in space at varying times. Following Batchelor (1967) and Gill (1982), the **conservation of mass**:

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}) = 0 \qquad \text{eq. 1-1}$$

while the **conservation of momentum**:

$$\frac{\partial \rho u}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}\boldsymbol{u}) + 2\boldsymbol{\Omega} \times (\rho \boldsymbol{u}) = -\boldsymbol{\nabla p} - \hat{\boldsymbol{k}}\boldsymbol{g}\rho + \boldsymbol{\nabla} \cdot (\mu \boldsymbol{\nabla u}) + \frac{1}{3}\boldsymbol{\nabla}(\mu \boldsymbol{\nabla} \cdot \boldsymbol{u}) \qquad \text{eq. 1-2}$$

In addition to the NS equation, in fluid dynamic, we consider the equation of state (EOS) relating the density to the thermodynamic variables:

$$\rho = f(T, S, p) \qquad \text{eq. 1-3}$$

and equations describing temperature and salinity conservation:

$$\frac{\partial \rho C}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u} C) = \boldsymbol{\nabla} \cdot (\rho k_c \boldsymbol{\nabla} C) \qquad \text{eq. 1-4}$$

The terminology here is standard, with **bold** letters indicating vectors, μ being the viscosity and k_c being the diffusivity of the generic tracer C. It should be noted that C is defined as the mass of tracer contained in unit mass of fluid (Gill, 1982). Here C can be interpreted as potential temperature, salinity, or a generic passive tracer.

1.1. From Navier-Stokes to Primitive Equations

1.1.1. Hydrostatic approximation

The hydrostatic approximation is a simplification of the equations governing the vertical component of velocity. It simply states that the pressure at any point in the ocean is due to the weight of the water above it. When vertical accelerations are small compared to the gravitational acceleration, the hydrostatic approximation is valid. Rewriting the vertical momentum equation in terms of scales of motions the only term that can balance the gravity is the pressure term, so the vertical component of the momentum equation in first order approximation can be simplified as:

$$\frac{\partial p}{\partial z} = -\rho g$$

In rare instances when disparity between horizontal and vertical scales does not exist, or when horizontal/vertical scales are particularly small, such as in convection plumes and short internal waves, the hydrostatic approximation ceases to hold, and the vertical-momentum balance must include all the terms.

1.1.2. Boussinesq approximation

The density variations in the ocean are quite small compared to the mean density, and we may exploit this to derive a simpler but still quite accurate equations of motion. The Boussinesq approximation states that the density variation is only important in the buoyancy term, ρg, and can be neglected in the rest of the equations. The Boussinesq approximation is thus valid if

$$\frac{\Delta\rho}{\rho} \ll 1$$

The conservation of momentum equation becomes:

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + 2\boldsymbol{\Omega} \times \boldsymbol{u} = -\frac{\delta\rho}{\rho_0}\boldsymbol{g}_e - \frac{1}{\rho_0}\boldsymbol{\nabla} p + \boldsymbol{\nabla} \cdot (\nu \boldsymbol{\nabla} \boldsymbol{u}) + \frac{1}{3}\boldsymbol{\nabla}(\nu \boldsymbol{\nabla} \cdot \boldsymbol{u})$$

Where $\nu = \frac{\mu}{\rho_0}$ and we substituted the hydrostatic balance for the reference pressure.

1.1.3. Incompressible approximation

For incompressible fluid, density along the line of flow remains constant over time,

$$\frac{D\rho}{Dt} = 0$$

Therefore, the continuity equation states that divergence of velocity is always zero:

$$\boldsymbol{\nabla} \cdot \vec{\boldsymbol{u}} = 0$$

Incompressibility rules out pressure waves like sound, so this simplification is not useful if these phenomena are of interest.

1.1.4. Reynolds averages

Consider splitting in two parts the total relative velocity field in the fluid, $\boldsymbol{u}$. The first, denoted by $\langle \boldsymbol{u} \rangle$, representing the large-scale flow we wish to describe in-detail. The second part, $\boldsymbol{u'}$, representing the smaller-scale turbulence, which is not directly of interest except insofar as it affects the large scale. To some degree $\boldsymbol{u'}$ is specified by the condition that times short compared to the characteristic evolution time of $\langle \boldsymbol{u} \rangle$ are still long enough to provide an adequate averaging interval for $\boldsymbol{u'}$, so that over those times the average of $\boldsymbol{u'}$ vanishes, i.e., that for

$$\boldsymbol{u} = \langle \boldsymbol{u} \rangle + \boldsymbol{u}'$$

The average of the small-scale velocity

$$\langle \boldsymbol{u}' \rangle = 0$$

Hence the bracket has the operational meaning in terms of averaging, so that

$$\langle \langle \boldsymbol{u} \rangle + \boldsymbol{u}' \rangle = \langle \boldsymbol{u} \rangle$$

By applying the decomposition to the x (zonal) momentum equation and taking the average:

$$\underbrace{\frac{\partial\langle u\rangle}{\partial t}}_{1}+\underbrace{\langle u\rangle\frac{\partial\langle u\rangle}{\partial x}+\langle v\rangle\frac{\partial\langle u\rangle}{\partial y}+\langle w\rangle\frac{\partial\langle u\rangle}{\partial z}}_{2}-\underbrace{2\Omega sin\theta\,\langle v\rangle}_{7}$$

$$=\underbrace{-\frac{1}{\rho_0}\frac{\partial\langle p\rangle}{\partial x}}_{8}+\underbrace{\nu\boldsymbol{\nabla}^{2}\langle u\rangle}_{10}\underbrace{-\frac{\partial\langle u'u'\rangle}{\partial x}-\frac{\partial\langle v'u'\rangle}{\partial y}-\frac{\partial\langle w'u'\rangle}{\partial z}}_{4}$$

The equation for $\langle u\rangle$ is written entirely in terms of the large-scale velocity, with the important exception of the three last terms on the right-hand side. Although u', v', and w' have zero average, the momentum flux of the fluctuations, which is quadratic in the fluctuation velocities, need not vanish when averaged. For example, the term

$$\langle v'u'\rangle\equiv-\frac{\tau_{xy}}{\rho}$$

represents an average flux of x-momentum due to the small-scale motions across a surface $y = y_0$=constant. The momentum flux due to the small-scale motion is, as far as the large-scale motion is concerned, equivalent to a stress τ_{xy} on the large-scale flow. These stresses are called the *Reynolds stresses*. They represent no new physical mechanism on the fundamental fluid-dynamical level; rather, they are the consequence of our decision to focus entirely on the dynamics of the "large" scale.

A particularly simple model of this type for the stresses- which also preserves the necessary symmetry- is:

$$\tau_{xx}=2\rho A_H\frac{\partial\langle u\rangle}{\partial x}$$

$$\tau_{yy}=2\rho A_H\frac{\partial\langle v\rangle}{\partial y}$$

$$\tau_{xy}=\tau_{yx}=\rho A_H\left(\frac{\partial\langle v\rangle}{\partial x}+\frac{\partial\langle u\rangle}{\partial y}\right)$$

$$\tau_{xz}=\tau_{zx}=\rho A_V\frac{\partial\langle u\rangle}{\partial z}+\rho A_H\frac{\partial\langle w\rangle}{\partial x}$$

$$\tau_{yz}=\tau_{zy}=\rho A_V\frac{\partial\langle v\rangle}{\partial z}+\rho A_H\frac{\partial\langle w\rangle}{\partial y}$$

The coefficients A_H, and A_V are called the horizontal and vertical turbulent viscosity coefficients respectively. The anisotropy of the large scale between the horizontal and vertical scales of the flow suggests that the mixing of largescale momentum in the two directions cannot be expected to be the same. We can write (using the incompressible version of the continuity equation to eliminate the cross-derivative terms):

$$\frac{\partial u}{\partial t} + \boldsymbol{u} \cdot \nabla u - 2\Omega \sin\theta \ v = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} + \nu \boldsymbol{\nabla}^2 u + \frac{1}{\rho}\nabla_h \cdot (A_H \nabla u) + \frac{1}{\rho}\frac{\partial}{\partial z}\left(A_V \frac{\partial u}{\partial z}\right)$$

$$\frac{\partial v}{\partial t} + \boldsymbol{u} \cdot \nabla v + 2\Omega \sin\theta \ u = -\frac{1}{\rho_0}\frac{\partial p}{\partial y} + \nu \boldsymbol{\nabla}^2 u + \frac{1}{\rho}\nabla_h \cdot (A_H \nabla v) + \frac{1}{\rho}\frac{\partial}{\partial z}\left(A_V \frac{\partial v}{\partial z}\right)$$

Where we omitted the average operator.

1.2. Temperature and Salinity equations

The equation governing temperature (T) arises from conservation of energy. The rate of change of T in time is the sum of a molecular diffusive term plus a term due to expansion/contraction in volume (potential temperature!).

For incompressible fluid

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + w\frac{\partial T}{\partial z} = \frac{k_T}{\rho C_v}\boldsymbol{\nabla}^2 T$$

Which is the advection diffusion equation.

While for the salt budget:

$$\frac{\partial S}{\partial t} + u\frac{\partial S}{\partial x} + v\frac{\partial S}{\partial y} + w\frac{\partial S}{\partial z} = k_S \boldsymbol{\nabla}^2 S$$

which simply states that a seawater parcel conserves its salt content, except for redistribution by diffusion. The coefficient k_S is the coefficient of salt diffusion, which plays a role analogous to the heat diffusivity k_T.

Temperature and Salinity equations can also be rewritten accounting for the scale separation introduced by Reynolds. We can re-write the equation for the “large” scale T and S as:

$$\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}=k_T\boldsymbol{\nabla}^2T+\frac{K_h}{\rho}\boldsymbol{\nabla}_h^2T+\frac{K_v}{\rho}\frac{\partial^2T}{\partial z^2}$$

$$\frac{\partial S}{\partial t}+u\frac{\partial S}{\partial x}+v\frac{\partial S}{\partial y}+w\frac{\partial S}{\partial z}=k_S\boldsymbol{\nabla}^2S+\frac{K_h}{\rho}\boldsymbol{\nabla}_h^2S+\frac{K_v}{\rho}\frac{\partial^2S}{\partial z^2}$$

Where $K_h$and K_v are the horizontal and vertical turbulent diffusivity coefficients respectively and the average operator has been omitted for simplicity.

In the temperature equation if we want to consider effects of light penetration (radiative absorption) a further term should be considered

$$\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}=k_T\boldsymbol{\nabla}^2T+\frac{K_h}{\rho}\boldsymbol{\nabla}_h^2T+\frac{K_v}{\rho}\frac{\partial^2T}{\partial z^2}+\frac{1}{\rho c_v}\frac{\partial I}{\partial z}$$

1.3. The Primitive Equations

Our set of governing equations is now complete. For *seawater* there are seven variables (*u, v, w, p,* ρ, *T* and *S*), for which we have a continuity equation, three momentum equations, an equation of state, a *temperature (or energy)* equation, and a *salinity* equation.

$$\frac{\partial u}{\partial t}+\boldsymbol{u}\cdot\nabla u-2\Omega\sin\theta\ v=-\frac{1}{\rho_0}\frac{\partial p}{\partial x}+\nu\boldsymbol{\nabla}^2u+\frac{1}{\rho}\nabla_h\cdot(A_H\nabla u)+\frac{1}{\rho_0}\frac{\partial}{\partial z}\left(A_V\frac{\partial u}{\partial z}\right)$$

$$\frac{\partial v}{\partial t}+\boldsymbol{u}\cdot\nabla v+2\Omega\sin\theta\ u=-\frac{1}{\rho_0}\frac{\partial p}{\partial y}+\nu\boldsymbol{\nabla}^2v+\frac{1}{\rho}\nabla_h\cdot(A_H\nabla v)+\frac{1}{\rho_0}\frac{\partial}{\partial z}\left(A_V\frac{\partial v}{\partial z}\right)$$

$$\frac{\partial p}{\partial z}=-\rho g$$

$$\boldsymbol{\nabla}\cdot\vec{\boldsymbol{u}}=0$$

$$\rho=f(T,S,p)$$

$$\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}=k_T\boldsymbol{\nabla}^2T+\frac{K_h}{\rho}\boldsymbol{\nabla}_h^2T+\frac{K_v}{\rho}\frac{\partial^2T}{\partial z^2}+\frac{1}{\rho c_v}\frac{\partial I}{\partial z}$$

$$\frac{\partial S}{\partial t}+u\frac{\partial S}{\partial x}+v\frac{\partial S}{\partial y}+w\frac{\partial S}{\partial z}=k_S\boldsymbol{\nabla}^2S+\frac{K_h}{\rho}\boldsymbol{\nabla}_h^2S+\frac{K_v}{\rho}\frac{\partial^2S}{\partial z^2}$$

2. BOUNDARY CONDITIONS

Navier-Stokes equations require the imposition of appropriate initial and boundary conditions to find a specific solution. The boundary conditions refer to the 'physical limits' of the fluid under consideration, and they are specific for every problem of interest. For the ocean there are several distinct physical limits:

1) the air-sea interface (vertical boundary),
2) the bathymetry (vertical boundary),
3) the coast (lateral boundary).

Those are kinematic boundary conditions. On the other hand, if we consider the interaction with the atmosphere we deal with "dynamic" boundary conditions. When we solve the Primitive equations (or the NS) numerically it may be necessary to limit the domain in specific areas, thus a new "artificial" boundary arises, which is the connection of the limited areas we are studying with the rest of the ocean. These boundaries are called "lateral open boundary".

2.1. Kinematic boundary conditions

As already anticipated, considering the horizontal boundaries of the ocean basins (the coasts) we assume first that it is rigid and therefore it is required that the normal velocity to the coasts is zero, namely:

$$\boldsymbol{u}_h \cdot \boldsymbol{n} = 0$$

where $\boldsymbol{n}$ is the unit vector normal to the coast $\boldsymbol{u}_h = (u, v)$ is the horizontal velocity vector. If the fluid is viscous, we must also impose that the velocity parallel to the boundary is zero, that is:

$$\boldsymbol{u}_h \times \boldsymbol{n} = 0$$

The above conditions are referred as "no-slip".

If we also consider a generic tracer C, such as salinity and temperature, we must impose additionally that the diffusive fluxes, defined as $\boldsymbol{J}_{diff} = K_h \boldsymbol{\nabla} C$ are zero in their normal component at the rigid boundary, that is:

$\boldsymbol{J}_{diff} \cdot \boldsymbol{n} = 0.$

For the boundary at the air-sea interface the situation is different because the boundary is not rigid, but it has a deformable free surface η. In general, we can write:

$$z = z_0 + \eta(x, y, t)$$

where z_0 it is the equilibrium position of the surface in the case of no motion.

A particle of fluid at the surface is required to remain there following the fluid motion, that is:

$$\frac{D}{Dt}(z - z_0 - \eta) = 0$$

which is equivalent to:

$$w|_{z=\eta} = \left(\frac{\partial \eta}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla}\eta\right)$$

On the ocean bathymetry, represented by the surface *H(x, y),* the condition is that a particle must not leave the bathymetry, that is:

$$\left(\frac{\partial}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla}\right)(z + H) = 0$$

$$w|_{z=-H(x,y)} = -\left(u\frac{\partial H}{\partial x} + v\frac{\partial H}{\partial y}\right) = -\boldsymbol{u}|_{z=-H(x,y)} \cdot \boldsymbol{\nabla}H$$

2.2. Dynamic Boundary Conditions

Dynamic boundary conditions consider the forcing of the ocean circulation at the air-sea interface. These conditions therefore include the main forcing of the ocean circulation, such as the heat, water and momentum fluxes and the river discharge. Among these forcing the main one is the moment flux which is related to the atmospheric wind.

Using the law of the wall, the wind stress is:

$$\tau_w^x = \rho_a C_D |\boldsymbol{u}_w| u_w$$

$$\tau_w^y = \rho_a C_D |\boldsymbol{u}_w| v_w$$

where (u_W, v_W) is the wind a 10m. At the interface, z = η, the turbulent stresses in the ocean must be set equal to the wind stress at the surface, that is:

$$A_V \frac{\partial \boldsymbol{u}}{\partial z}\bigg|_{z=\eta} = \boldsymbol{\tau}_w$$

The momentum flux at the air-sea interface is equal to the turbulent flux in the water column.

At the water-sediment interface or the bathymetry, we define, in analogy with the law of the wall for the air-sea interface, the bottom stress:

$$\tau_b^x = \rho_o C_b |\boldsymbol{u}_b| u_b$$

$$\tau_b^y = \rho_o C_b |\boldsymbol{u}_b| v_b$$

where $\boldsymbol{u}_b$ is the velocity close to the bottom and C_b is the bottom drag coefficient defined in a similar way to that of the wind but with a different interpretation of the roughness length that this time is due to ripples on the bottom. The dynamic boundary condition at the bathymetry is then:

$$A_V \frac{\partial \boldsymbol{u}}{\partial z}\bigg|_{z=-H} = \boldsymbol{\tau}_b$$

The boundary conditions for the temperature are derived from the same concept of equality between the heat fluxes at the air-sea interface and the turbulent heat fluxes in the water column. It is therefore written:

$$K_v \frac{\partial T}{\partial z}\bigg|_{z=\eta} = \frac{Q_t}{C_p \rho_0}$$

where Q_t it is the heat exchanged with the atmosphere, positive when is downward, C_p the specific heat and ρ_0 the density of seawater. Two fundamental processes take place at the air-sea interface:

- The first is characterised by the radiative exchange of heat, the absorption of solar incident radiation and its re-emission by the sea surface and the atmosphere (greenhouse effect). This is called overall the radiative balance at the air-sea interface.

– The second is the turbulent flux due to conduction/convention, dry and moist, with release of sensible and latent heat respectively.

The overall heat flux at the air-sea interface can be written as:

$$Q_t = Q_s - Q_b - Q_h - Q_e$$

where: Q_s is the incident solar radiation, Q_b the radiation re-emitted by the sea, Q_h is the sensible heat and Q_e the latent heat flux. The units are W m^{-2}. Example bulk formulae are provided in the following.

Q_s is proportional to the incident solar radiation at short wavelength (visible and near infrared), to the cloudiness and to the albedo:

$$Q_s = f(S, \alpha, c) \approx \frac{S(1-\alpha)}{4}(1-0.7c)$$

where c is the fraction of sky covered by clouds and α is the albedo.

Q_b is given by the net balance between the radiation emitted by the sea and the infrared radiation that comes from the atmosphere. The radiation emitted from the surface is calculated using the Stefan-Boltzmann law, $E = \sigma T_o^4$, where σ is the Stephan-Boltzmann constant and T_o is the temperature of the sea. The infrared radiation that returns to the surface due to the greenhouse effect is a complicated function in the first instance of the chemical composition of the atmosphere and the clouds. An example of such a formula is:

$$Q_b = f(T_o^4, e_a, c) \approx 0.985\sigma T_o^4 \left(0.39 - 0.05 e_a^{\frac{1}{2}}\right)(1 - 0.5c^2)$$

where e_a it is the water vapor pressure at the surface considering the air to be saturated at the temperature T_o.

As for sensible heat, in the first 10 meters layer above the surface, the heat released for turbulent processes it is proportional to the speed of the fluid and the difference of temperature between the ocean and the atmosphere:

$$Q_h = \rho_a C_p C_h |\boldsymbol{u}| (T_o - T_a)$$

where C_h is the Stanton coefficient, C_p it is the specific heat at constant pressure and ρ_a is the density of the air.

A possible formulation of latent heat fluxes is:

$$Q_e = \rho_a L C_e |\boldsymbol{u}_b| (q_s - q_a)$$

where C_e it is called the Danton coefficient and L it is the latent heat of evaporation $q_{s,a}$ are the specific humidity at 10m and of sea surface at saturation temperature T_o.

Regarding the exchange of water at the air-sea interface, the water flux is:

$$q_w = E - P - \frac{R}{A}$$

where E is the evaporation, P is the precipitation, both in units of [m s^{-1}], R the river discharge in [m^3 s^{-1}] and A it is the area of the section at the mouth of the river. In this formulation the exchange with the rivers is supposed to occur entirely at the surface of the sea which is an approximation commonly used in large scale ocean modelling. In the reality, the river contribution extends deeper in the water column and depends on the type of estuary or delta in question. The condition on water exchange modifies the kinematic boundary condition which then becomes:

$$w|_{z=\eta} = \left(\frac{\partial \eta}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \eta\right) + q_w$$

If the water flux is different from zero, the salinity must change for the salinity to be conserved in a closed basin. The boundary condition for salinity is therefore:

$$K_v \frac{\partial S}{\partial z}\bigg|_{z=\eta} = \left(E - P - \frac{R}{A}\right) S_0$$

where S_0 is the salinity on the surface.

3. TAYLOR SERIES AND FINITE DIFFERENCES

To numerically solve continuous differential equations, we must first define how to represent a continuous function by a finite set of numbers, f_j with j =1, 2, ..., and then translate the continuous differential equations into a finite set of algebraic equations that determine the values, f_j. Two basic strategies exist: the grid-point methods (finite difference and finite volume); and series expansion methods (spectral, finite element, ...). Of grid-point methods, the finite difference method is the most widely used in ocean modelling and weather forecasting but in recent years, the finite volume method gained some popularity. Spectral methods are widely used in meteorology because they are both accurate and efficient, but they are rarely considered in oceanography.

3.1. Representation of functions

In Fig.1, a continuous and differentiable function, f, of the independent variable x, is approximated by the values $f_i = f(x_i)$ at the points, x_i. The set of points, x_i, make up the numerical mesh or "grid". The spacing between the grid points need not be constant; if the spacing is constant then the grid is regular.

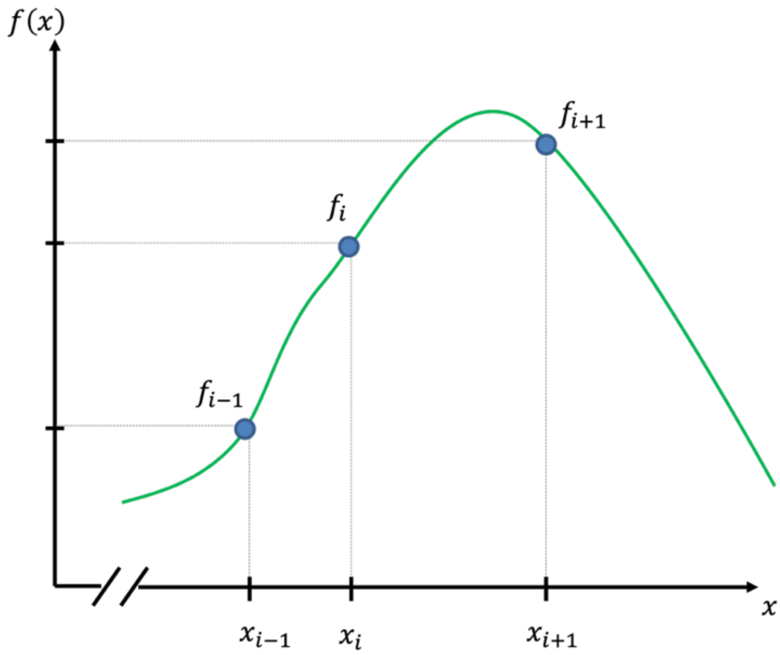


Figure 1 *An approximation to a smooth and continuous function, f(x), can be represented by the truncated set of values {f_{i-1},f_i,...} evaluated at the grid points {x_{i-1},x_i,...} where* $\boldsymbol{f_i \equiv f(x)}$

No assumption is made about the value of the approximate solution between the grid points. However, in the space between discrete grid points, for example $x_{i-1} < x' < x_i$, the function can be interpolated assuming a particular representation of the function based on the known values f_{i-1}, f_i, ... For example, Fig.2 shows a straight line drawn between values f_{i-1} and f_i which is given by:

$$\tilde{f}(x') = \left(1 - \alpha(x')\right) f_{i-1} + \alpha(x') f_i$$

$$\alpha(x') \equiv \frac{x' - x_{i-1}}{x_i - x_{i-1}} \qquad \forall\, x_{i-1} \le x' \le x_i$$

so that $\tilde{f}(x')$ is the approximation to $f(x)$. This is "linear interpolation". Higher order interpolation and other representations (such as quadratic, spline or high order polynomial) are possible.

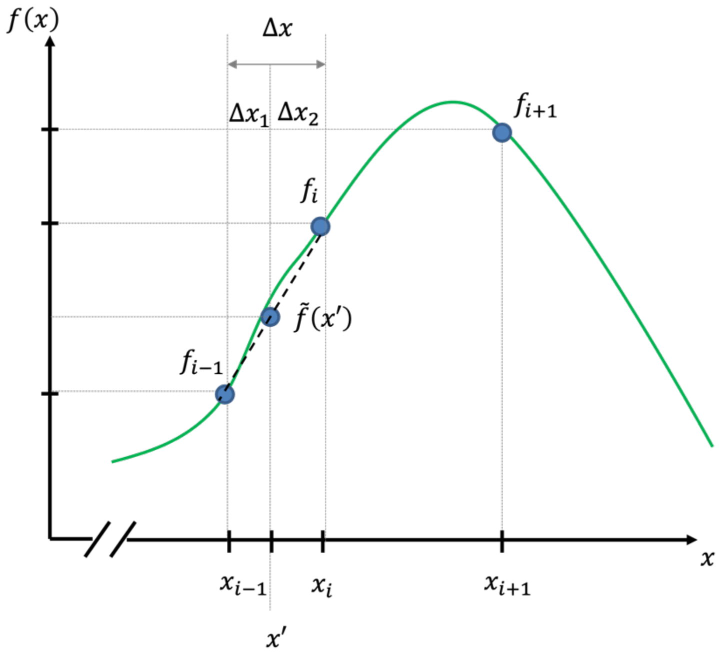


***Figure 2** Linear interpolation (dashed line) between values f_i of a function $f(x)$ (thin line) on a discrete grid x_i.*

3.2. Taylor series and truncation error

Taylor series can be used to estimate a truncation error in an approximation of a function and used to establish approximations to derivatives. The Taylor series expansion at a point *x* is:

$$f(x \pm \epsilon) = f(x) + \sum_{j=1}^{\infty} \epsilon^j \frac{(\pm 1)^j}{j!} \frac{\partial^j}{\partial x^j} f$$

or:

$$f(x \pm \epsilon) = f(x) \pm \epsilon f'(x) + \frac{\epsilon^2}{2!} f''(x) \pm \frac{\epsilon^3}{3!} f'''(x) + \frac{\epsilon^4}{4!} f''''(x) \pm \ldots + \ldots \pm$$

where ϵ is some small displacement from *x*.

To estimate the error incurred evaluating the function *f(x)* using linear interpolation we can replace the values f_{i-1} and f_i with Taylor series expansions about the point of evaluation, *x*:

$$f_{i-1} = f(x - \Delta x_1) = f(x) - \Delta x_1 f'(x) + \frac{{\Delta x_1}^2}{2!} f''(x) - \ldots$$

$$f_i = f(x + \Delta x_2) = f(x) + \Delta x_2 f'(x) + \frac{{\Delta x_2}^2}{2!} f''(x) - \ldots$$

Where $\Delta x_1 \equiv x - x_{i-1}$ and $\Delta x_2 \equiv x_i - x$.

Substituting gives

$$\begin{aligned}\tilde{f}(x) &= \frac{\Delta x_2}{\Delta x_1 + \Delta x_2}\left[f(x) - \Delta x_1 f'(x) + \frac{1}{2}\Delta x_1^2 f''(x) - \ldots\right] \\ &\quad + \frac{\Delta x_1}{\Delta x_1 + \Delta x_2}\left[f(x) + \Delta x_2 f'(x) + \frac{1}{2}\Delta x_2^2 f''(x) + \ldots\right] \\ &= f(x) + \frac{1}{2}\Delta x_1 \Delta x_2 f''(x) + \ldots\end{aligned}$$

This allows us to estimate the error $\tilde{f}(x) - f(x)$ as

$$\tilde{f}(x) - f(x) = \frac{1}{2}\Delta x_1(\Delta x - \Delta x_1) f''(x) = \frac{1}{2}\Delta x^2 \gamma(1-\gamma) f''(x)$$

dropping higher order terms. Here $\Delta x = x_i - x_{i-1} = \Delta x_1 + \Delta x_2$ is the grid spacing and

$$\gamma = \frac{\Delta x_1}{\Delta x} = \frac{\Delta x_1}{\Delta x_1 + \Delta x_2}$$

is the non-dimensional position within the space interval ($0 \leq \gamma \leq 1$). Note some properties about the estimate of error for linear interpolation:

- the error scales with the square of the grid spacing (Δx^2),
- the error is reduced if the interpolation point is near the ends of the interval ($\gamma^2 \sim 0,1$). i.e. near a data value,
- the error is largest in the middle of the interval ($\gamma \sim \frac{1}{2}$), equidistant from the data values.

The order of accuracy of an approximation refers to the scaling with mesh parameters (normally the grid resolution). The above analysis tells us that linear interpolation is second order accurate, or that errors are of order $O(\Delta x^2)$. This means that doubling the resolution (i.e. halving the grid spacing) reduces the error by a factor of four.

3.3. Constructing difference operators using Taylor series

Finite differencing is the method of approximating partial derivatives with differences between function values (on a grid). The Taylor series for $f_{i-1}(x)$, $f_i(x)$,... can be re-arranged to yield expressions for the derivatives, $f'(x)$,$f''(x)$,... yielding formula for approximating derivatives with discrete differences and truncation error.

As an example, consider how to estimate the first derivative, f', at x_i in terms of f_{i-1} and f_i. The Taylor series for f_{i-1} and f_i expanded about x_i are:

$$f_{i-1} = f(x_i) - \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) - \ldots$$

$$f_i = f(x_i)$$

Where $\Delta x = x_i - x_{i-1}$ is the grid spacing. Solving for $f'(x_i)$ gives:

$$f'(x_i) \approx \frac{1}{\Delta x}(f_i - f_{i-1}) + \frac{\Delta x}{2} f''(x_i)$$

where we have dropped higher order terms. This difference formula is known as a side difference and is first order accurate, i.e. the truncation error is of order $O(\Delta x)$. This is the simplest and least accurate of the finite difference approximations.

Now, consider the same data points but this time evaluate the derivative at the mid-point, $\bar{x} = \frac{1}{2}(x_{i-1} + x_i)$. The Taylor series are:

$$f_{i-1} = f(\bar{x}) - \frac{\Delta x}{2} f'(\bar{x}) + \frac{1}{2!}\left(\frac{\Delta x}{2}\right)^2 f''(\bar{x}) - \frac{1}{3!}\left(\frac{\Delta x}{2}\right)^3 f'''(\bar{x}) + \frac{1}{4!}\left(\frac{\Delta x}{2}\right)^4 f''''(\bar{x}) - \ldots$$

$$f_i = f(\bar{x}) + \frac{\Delta x}{2} f'(\bar{x}) + \frac{1}{2!}\left(\frac{\Delta x}{2}\right)^2 f''(\bar{x}) + \frac{1}{3!}\left(\frac{\Delta x}{2}\right)^3 f'''(\bar{x}) + \frac{1}{4!}\left(\frac{\Delta x}{2}\right)^4 f''''(\bar{x}) - \ldots$$

Solving for $f'(\bar{x})$ gives:

$$f'(\bar{x}) \approx \frac{1}{\Delta x}(f_i - f_{i-1}) + \frac{\Delta x^2}{4 \cdot 3!} f'''(\bar{x})$$

This is a centered difference and is second order accurate because the truncation error is of order $O(\Delta x^2)$. Because the evaluation point is at the mid-point between grid nodes, the derivative is said to be staggered in space.

Comparing the two formulae, we see that the difference operator is the same in each but the point at which the derivative is being approximated matters. Although the centered difference is formally of higher order the truncation terms cannot be directly compared since $f''(x_i)$ and $f'''(x_i)$ are different and not known.

As a rule, centered evaluations of difference formula are more accurate than off-center or side differences. Further, centered differences are generally of even order and side differences typically of odd order, though it is possible to construct even order side differences by placing all points of the stencil to one side of the evaluation point.

Finally, consider a centered difference evaluated at a grid point (rather than a mid-point). Limiting ourselves to a uniform grid (i.e. $x_{i+1} - x_i = x_i - x_{i-1} = \Delta x$) we can construct the operator by applying the formula over a 2Δx interval, yielding:

$$f'(x_i) \approx \frac{1}{2\Delta x}(f_{i+1} - f_{i-1}) + \frac{\Delta x^2}{3!} f'''(x_i)$$

This is also of second order accuracy but here the coefficient in the truncation error is 4 times larger than for the staggered center difference. Therefore, although we cannot directly compare $f'''(x_i)$ with $f'''(\bar{x})$ because they are at different locations, we can infer that the staggered difference is more accurate than the 2Δx difference when the variations in the function are well resolved.

3.3.1. Second derivative

To find a formula for the second derivative, *f''*, we must use a stencil of at least 3 points. Here, we'll consider a uniform grid with spacing Δx. The Taylor series expanded about x_i are:

$$f_{i-1} = f(x_i) - \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) - \frac{\Delta x^3}{3!} f'''(x_i) + \frac{\Delta x^4}{4!} f''''(x_i) - \dots$$

$$f_i = f(x_i)$$

$$f_{i+1} = f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + \frac{\Delta x^3}{3!} f'''(x_i) + \frac{\Delta x^4}{4!} f''''(x_i) - \dots$$

Solving for $f''(x_i)$ and dropping higher order terms gives:

$$f''(x_i) \approx \frac{f_{i+1} - 2f_i + f_{i-1}}{\Delta x^2} - \frac{2\Delta x^2}{4!} f''''(x_i)$$

3.3.2. Generating higher order differences

The process of finding difference formula and truncation terms involves several expansions of the Taylor series about the same point and then solving the resulting equations for the unknown derivatives. This can be cast as a linear algebra problem where we can use Gaussian elimination to solve for the derivatives.

The three Taylor series used in previous section can be written as a matrix problem:

$$
\left(\begin{array}{ccc|cc}
1 & -\Delta x & \dfrac{\Delta x^2}{2} & -\dfrac{\Delta x^3}{3!} & \dfrac{\Delta x^4}{4!} \\
1 & 0 & 0 & 0 & 0 \\
1 & \Delta x & \dfrac{\Delta x^2}{2} & \dfrac{\Delta x^3}{3!} & \dfrac{\Delta x^4}{4!}
\end{array}\right)
\begin{pmatrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f'''(x_i) \\ f''''(x_i) \end{pmatrix}
=
\begin{pmatrix} f_{i-1} \\ f_i \\ f_{i+1} \end{pmatrix}
$$

where the vertical line indicates a square sub-matrix. If we multiply through by the inverse of this sub-matrix then we get:

$$
\left(\begin{array}{ccc|cc}
1 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & \dfrac{\Delta x^2}{6} & 0 \\
0 & 0 & 1 & 0 & \dfrac{\Delta x^2}{12}
\end{array}\right)
\begin{pmatrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f'''(x_i) \\ f''''(x_i) \end{pmatrix}
=
\begin{pmatrix}
0 & 1 & 0 \\
-\dfrac{1}{2\Delta x} & 0 & \dfrac{1}{2\Delta x} \\
\dfrac{1}{\Delta x^2} & -\dfrac{2}{\Delta x^2} & \dfrac{1}{\Delta x^2}
\end{pmatrix}
\begin{pmatrix} f_{i-1} \\ f_i \\ f_{i+1} \end{pmatrix}
$$

This approach is identical to the derivations used in previous sections but in just one operation yields all the difference approximations of derivatives up to the order of the number of points.

Using five points, centered at x_i, the result is:

$$
\left(\begin{array}{ccccc|cc}
1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & -\dfrac{\Delta x^4}{30} & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & -\dfrac{\Delta x^4}{90} \\
0 & 0 & 0 & 1 & 0 & \dfrac{\Delta x^2}{4} & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & \dfrac{\Delta x^2}{6}
\end{array}\right)
\begin{pmatrix} f(x_i) \\ f' \\ f'' \\ f''' \\ f'''' \\ f''''' \\ f'''''' \end{pmatrix}
$$

$$
=
\begin{pmatrix}
0 & 0 & 1 & 0 & 0 \\
\dfrac{1}{12\Delta x} & -\dfrac{2}{3\Delta x} & 0 & \dfrac{2}{3\Delta x} & -\dfrac{1}{12\Delta x} \\
-\dfrac{1}{12\Delta x^2} & \dfrac{4}{3\Delta x^2} & -\dfrac{5}{2\Delta x^2} & -\dfrac{4}{3\Delta x^2} & -\dfrac{1}{12\Delta x^2} \\
-\dfrac{1}{2\Delta x^3} & -\dfrac{1}{\Delta x^3} & 0 & -\dfrac{1}{\Delta x^3} & -\dfrac{1}{2\Delta x^3} \\
\dfrac{1}{12\Delta x^4} & -\dfrac{4}{\Delta x^4} & \dfrac{6}{\Delta x^4} & -\dfrac{4}{\Delta x^4} & \dfrac{1}{\Delta x^4}
\end{pmatrix}
\begin{pmatrix} f_{i-2} \\ f_{i-1} \\ f_i \\ f_{i+1} \\ f_{i+2} \end{pmatrix}
$$

This gives us fourth order accurate formula for $f'(x)$ and $f''(x)$ and second order accurate formula for $f'''(x)$ and $f''''(x)$.

3.4. Some commonly used "finite difference" formulae.

First derivative

$f'(x_i) \approx \frac{f(x_{i+1})-f(x_i)}{\Delta x} + O(\Delta x)$ Forward

$f'(x_i) \approx \frac{f(x_i)-f(x_{i-1})}{\Delta x} + O(\Delta x)$ Backward

$f'(x_i) \approx \frac{f(x_{i+1})-f(x_{i-1})}{2\Delta x} + O(\Delta x^2)$ Centered

Second derivative

$f''(x_i) \approx \frac{f(x_{i+2})-2f(x_{i+1})+f(x_i)}{\Delta x^2} + O(\Delta x)$ Forward

$f''(x_i) \approx \frac{f(x_i)-2f(x_{i-1})+f(x_{i-2})}{\Delta x^2} + O(\Delta x)$ Backward

$f''(x_i) \approx \frac{f(x_i+1)-2f(x_i)+f(x_{i-1})}{\Delta x^2} + O(\Delta x^2)$ Centered

4. SPACE AND TIME INTEGRATION

4.1. The heat equation.

A simple, but revealing, physical equation which emphasizes the differences among the various approaches is the one-dimensional heat equation:

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2} \qquad 0 \leq x \leq L, \qquad t \geq 0$$

This is a parabolic mixed initial/boundary value problem requiring boundary conditions in both space and time. Simple boundary and initial conditions are:

$$\phi(0,t) = \phi_0, \qquad \phi(L,t) = \phi_L, \qquad \phi(x,0) = f_0(x)$$

The finite difference approximations developed in the preceding section are now assembled into a discrete approximation to heat equation. Both the time and space derivatives are replaced by finite differences. Doing so requires specification both the time and spatial locations of the ϕ values in the finite difference formulas. Therefore, we need to introduce superscript m to designate the time step of the discrete solution. Though this seems like only is a bookkeeping issue, we shall soon see that choosing the time step at which the spatial derivatives are evaluated will have a large impact on the performance and ease of implementation of the finite difference model.

4.2. Forward Time, Centered Space

Approximate the time derivative in the heat equation with a forward difference:

$$\left.\frac{\partial \phi}{\partial t}\right|_{t_{m+1}, x_i} \approx \frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} + O(\Delta t)$$

Note that terms on the right-hand side only involve ϕ at $x = x_i$.

Use the central difference approximation to $\left(\frac{\partial^2 \phi}{\partial x^2}\right)_{x_i}$ and evaluate all terms at time m.

$$\left.\frac{\partial^2 \phi}{\partial x^2}\right|_{x_i} \approx \frac{\phi_{i-1}^m - 2\phi_i^m + \phi_{i+1}^m}{\Delta x^2} + O(\Delta x^2)$$

Substitute into the heat equation, collect the truncation error terms to get:

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} = K \frac{\phi_{i-1}^m - 2\phi_i^m + \phi_{i+1}^m}{\Delta x^2} + O(\Delta t) + O(\Delta x^2)$$

The temporal errors and the spatial errors have different orders. Also notice that we can explicitly solve for ϕ_i^{m+1} in terms of the other values of ϕ. Drop the truncation error terms and solve for ϕ_i^{m+1} to get:

$$\phi_i^{m+1} = \phi_i^m + \frac{K\Delta t}{\Delta x^2}(\phi_{i-1}^m - 2\phi_i^m + \phi_{i+1}^m)$$

This equation is called the Forward Time, Centered Space or FTCS approximation to the heat equation. A slight improvement in computational efficiency can be obtained with a small rearrangement:

$$\phi_i^{m+1} = r\phi_{i+1}^m + (1 - 2r)\phi_i^m + r\phi_{i-1}^m$$

where $r = \frac{K\Delta t}{\Delta x^2}$.

The FTCS scheme is easy to implement because the values of ϕ_i^{m+1} can be updated independently of each other. The entire solution is contained in two loops: an outer loop over all time steps, and an inner loop over all interior nodes.

FTCS equation can be represented by the stencil in the left third of Fig.3. Notice that the value of ϕ_i^{m+1} does not depend on ϕ_{i-1}^{m+1} or ϕ_{i+1}^{m+1} for the FTCS scheme.

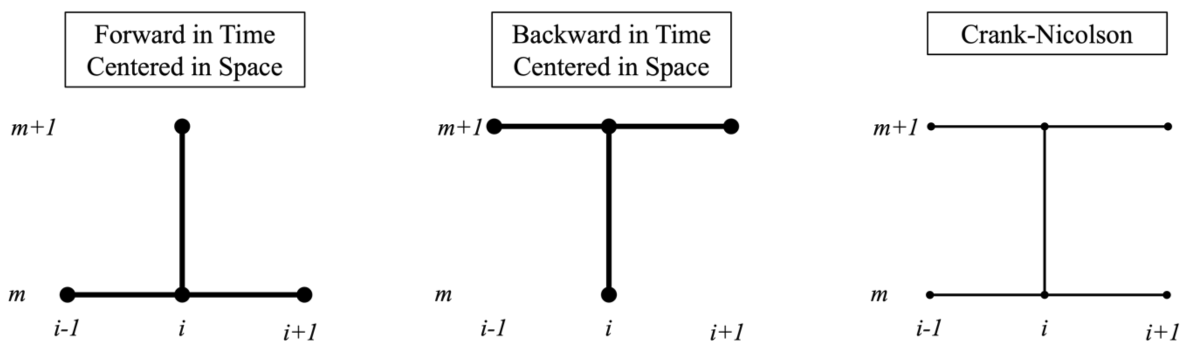


Figure 3

The solutions to Heat Equation subject to the initial and boundary conditions are all bounded, decaying functions. In other words, the magnitude of the solution will decrease from the initial condition to a constant. The FTCS can yield unstable solutions that oscillate and grow if Δt is too large. Stable solutions with the FTCS scheme are only obtained if

$r = \frac{K\Delta t}{\Delta x^2} < \frac{1}{2}$.

Finally, we note that

$$\phi_i^{m+1} = r\phi_{i+1}^m + (1-2r)\phi_i^m + r\phi_{i-1}^m$$

can be expressed as a matrix multiplication.

$$\phi^{m+1} = A\phi^m$$

where A is the tridiagonal matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ r & (1-2r) & r & 0 & 0 & 0 \\ 0 & r & (1-2r) & r & 0 & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & r & (1-2r) & r \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$\phi^{m+1}$is the vector of ϕ values of at time step $m + 1$, and ϕ^m is the vector of ϕ values at time step m. The first and last rows of A are adjusted so that the boundary values of ϕ are not changed when the matrix-vector product is computed.

4.3. Backward Time, Centered Space

In the derivation of the discretized Heat Equation to approximate the time derivative now, choose the backward difference:

$$\left.\frac{\partial\phi}{\partial t}\right|_{t_{m+1},x_i} \approx \frac{\phi_i^m - \phi_i^{m-1}}{\Delta t} + O(\Delta t)$$

Thus:

$$\frac{\phi_i^m - \phi_i^{m-1}}{\Delta t} = K\frac{\phi_{i-1}^m - 2\phi_i^m + \phi_{i+1}^m}{\Delta x^2} + O(\Delta t) + O(\Delta x^2)$$

The truncation errors in this approximation have the same order of magnitude as the truncation errors in FTCS. Unlike FTCS, however, the new equation cannot be rearranged to obtain a simple algebraic formula for computing for ϕ_i^min terms of its neighbours ϕ_{i+1}^m, , ϕ_{i-1}^m and ϕ_i^{m-1}. Developing a similar equation for ϕ_{i+1}^m shows that ϕ_{i+1}^m depends on $\phi_{i+2}^m$and ϕ_i^m. Thus, this is one equation in a system of equations for the values of ϕ at the internal nodes of the spatial mesh (i = 2, 3, . . ., N − 1). To see the system of equations more clearly, drop the truncation error terms and rearrange:

$$-\frac{K}{\Delta x^2}\phi_{i-1}^m + \left(\frac{1}{\Delta t} + \frac{2K}{\Delta x^2}\right)\phi_i^m - \frac{K}{\Delta x^2}\phi_{i+1}^m = \frac{1}{\Delta t}\phi_i^{m-1}$$

The system of equations can be represented in matrix form as

$$\begin{pmatrix} b_1 & c_1 & 0 & 0 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & 0 & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & 0 & 0 & 0 & a_N & b_N \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \vdots \\ \phi_{N-1} \\ \phi_N \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{pmatrix}$$

where the coefficients of the interior nodes are

$$a_i = -\frac{K}{\Delta x^2}$$

$$b_i = \frac{1}{\Delta t} + \frac{2K}{\Delta x^2}$$

$$c_i = -\frac{K}{\Delta x^2}$$

$$d_i = \frac{1}{\Delta t}\phi_i^{m-1}$$

For the Dirichlet (specify fixed values of ϕ at the boundaries) boundary conditions:

$b_1 = 1$, $c_1 = 0$, $d_1 = \phi_o$ $a_N = 0$, $b_N = 1$, $d_N = \phi_L$

Implementation of the BTCS scheme requires solving a system of equations at each time step. In addition to the complication of developing the code, the computational effort per time step for the BTCS scheme is larger than in the FTCS scheme. The BTCS scheme has one huge advantage over the FTCS: it is unconditional stable (in this case). The BTCS scheme is just as accurate as the FTCS scheme. Therefore, with some extra effort, the BTCS scheme yields a computational model that is robust to choices of Δt and Δx. This advantage is not always overwhelming, however, and the FTCS scheme is still useful for some problems.

4.4. Trapezoidal / Crank-Nicolson

Both the FTCS and the BTCS have a temporal truncation error of $O(\Delta t)$. When time-accurate solutions are important, the Trapezoidal scheme has significant advantages. The scheme is not significantly more difficult to implement than the BTCS scheme, and it has a temporal truncation error $O(\Delta t^2)$. The scheme is implicit, and it is unconditional stable. The scheme is equivalent to an implicit mid-point method.

The forward and backward methods can be written:

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} = F_i^m$$

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} = F_i^{m+1}$$

While the Crank-Nicolson method is:

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} = \frac{1}{2}\left(F_i^{m+1} + F_i^m\right)$$

The r.h.s of the heat equation is approximated with the average of the central difference scheme evaluated at the current and the previous time step. Thus, the heat equation is approximated with:

$$\frac{\phi_i^m - \phi_i^{m-1}}{\Delta t} = \frac{K}{2}\left(\frac{\phi_{i-1}^m - 2\phi_i^m + \phi_{i+1}^m}{\Delta x^2} + \frac{\phi_{i-1}^{m-1} - 2\phi_i^{m-1} + \phi_{i+1}^{m-1}}{\Delta x^2}\right)$$

Rearranging:

$$-\frac{K}{2\Delta x^2}\phi_{i-1}^{m}+\left(\frac{1}{\Delta t}+\frac{K}{\Delta x^2}\right)\phi_i^m-\frac{K}{2\Delta x^2}\phi_{i+1}^{m}$$
$$=\frac{K}{2\Delta x^2}\phi_{i+1}^{m-1}+\left(\frac{1}{\Delta t}+\frac{K}{\Delta x^2}\right)2\phi_i^{m-1}+\frac{K}{2\Delta x^2}\phi_{i+1}^{m-1}$$

The system of equation is identical to the BTCS but with different coefficients:

$$a_i=-\frac{K}{2\Delta x^2}$$

$$b_i=\frac{1}{\Delta t}+\frac{K}{\Delta x^2}$$

$$c_i=-\frac{K}{2\Delta x^2}$$

$$d_i=\frac{1}{\Delta t}\phi_i^{m-1}-a_i\phi_{i-1}^{m-1}+(a_i+c_i)\phi_i^{m-1}-c_i\phi_{i+1}^{m-1}$$

4.5. Examples

Integrating the heat equation with the 3 methods. We set a simple problem where analytical solution exists. Initial condition is:

$$f_0(x)=sin\left(\frac{\pi x}{L}\right)$$

And boundary conditions are:

$$\phi(0,t)=\phi(L,t)=0$$

The exact solution is:

$$\phi(x,t)=sin\left(\frac{\pi x}{L}\right)e^{\left(-\frac{k\pi^2 t}{L^2}\right)}$$

First test.

Assume the spatial dimension L from 0 to 1 (L), discretized using 20 points (nx), so that $\Delta x=0.0526$, the time dimension to go from 0 to 0.5 (tmax), discretized in 10 time-steps, so that $\Delta t=0.0556$ and the diff. coeff K=0.1.

We can perform the simulations with the 3 methods and quantify the error at the end of the simulation comparing the numerical and the analytical results.

FTCS error = 2.404e-02

BTCS error = 2.676e-02

CN error = 1.883e-03

Note that the norm of the error for the FTCS scheme and BTCS scheme are comparable: 0.024 and 0.027. The error of the Crank-Nicolson scheme is 0.0018, which is more than an order of magnitude smaller than either the FTCS and the BTCS schemes.

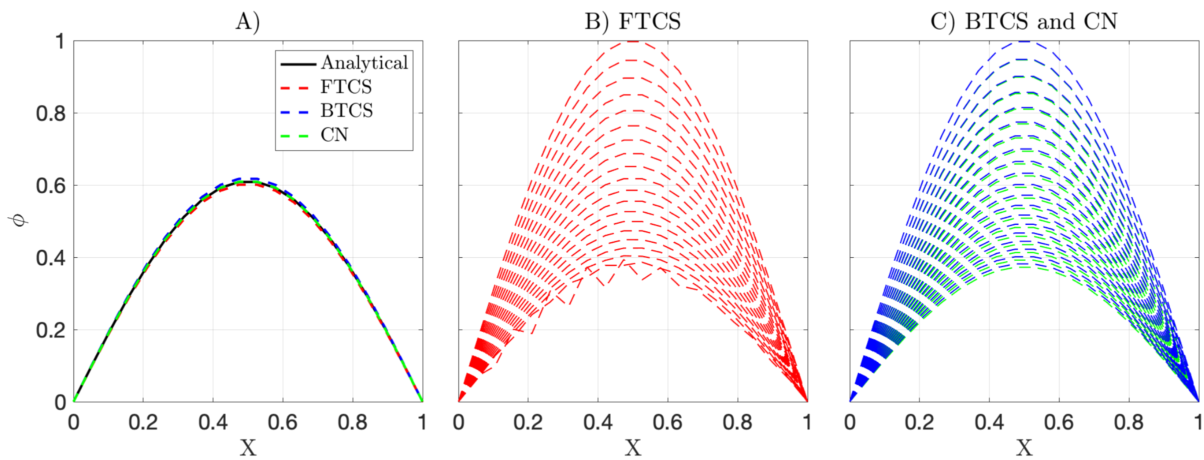


Figure 4 A) Comparison of the three solutions with the analytical solution at the end of the simulation. B) evolution in time of the FTCS solution for longer integration time (ϕ decreases with time). C) evolution in time of the BTCS and CN as panel B.

The heat equation is a model of diffusive systems. For all problems with constant boundary values, the solutions decay from an initial state to a steady state condition. These solutions are smooth and bounded: the solution does not develop local or global maxima that are outside the range of the initial data. However, we stated that the FTCS scheme can exhibit instability, so that the values of ϕ at the nodes may oscillate and grow outside the bounds of the initial and boundary conditions. We also stated the stability limit of the parameter r to be 0.5. In our setup r = 2, is significantly higher than the stability limit. We do not see any instability simply because the simulation is too short. Integrating longer in time using the same time-step we see that the solution starts to oscillate generating local maxima and minima, while the BTCS and the CN schemes they are both stable over time (Figure 4 panel B and C).

In Figure 5 the error as function of the time-step fixing the grid spacing in the different scheme. We can see that in the FTCS case when the stability factor becomes larger than 0.5 the error grows very rapidly. In the last panel of Figure 5 we can see the error growth for the CN and BTCS schemes for larger time-steps (where the FTCS is unstable).

Finally in Figure 6 the error as function of the spatial grid size is drown for the three schemes.

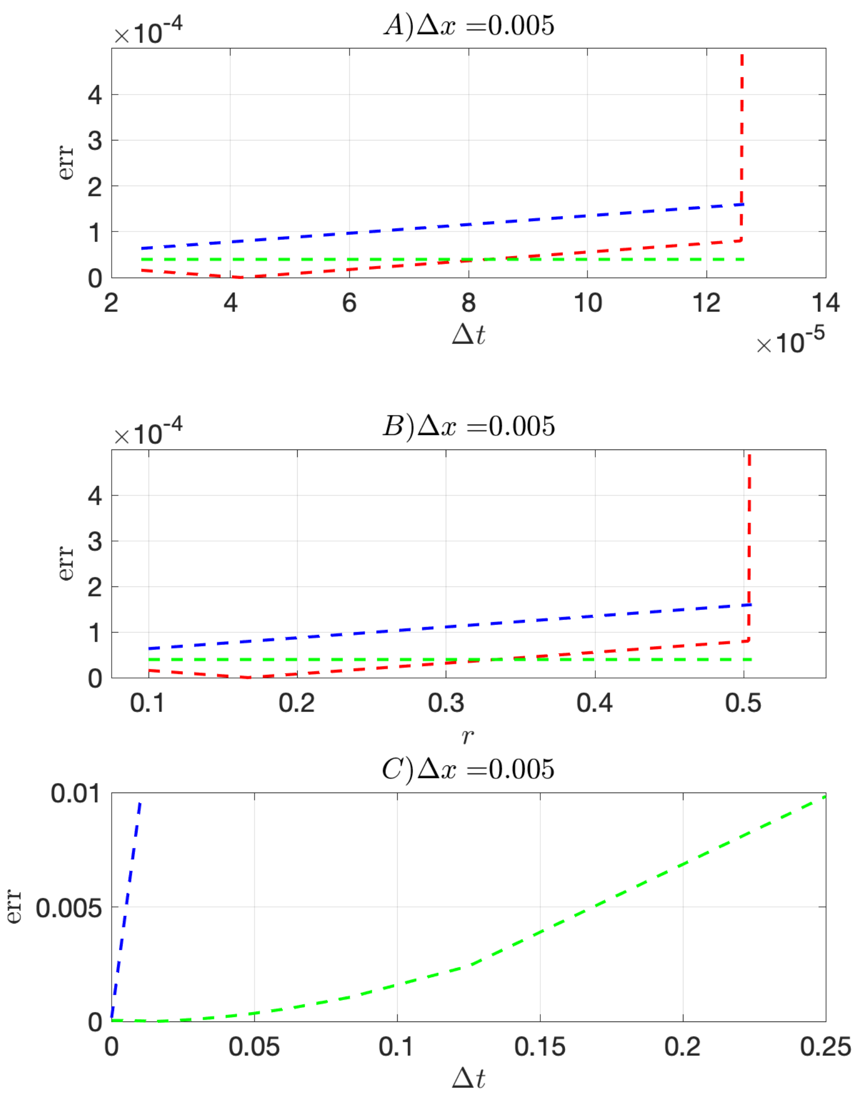


Figure 5 Error with fixed delta-x and variable delta-t. Red FTCS, blue BTCS, green CN. A) as function of time-step size, B) as function of the stability parameter (r), D) for larger time-step size.

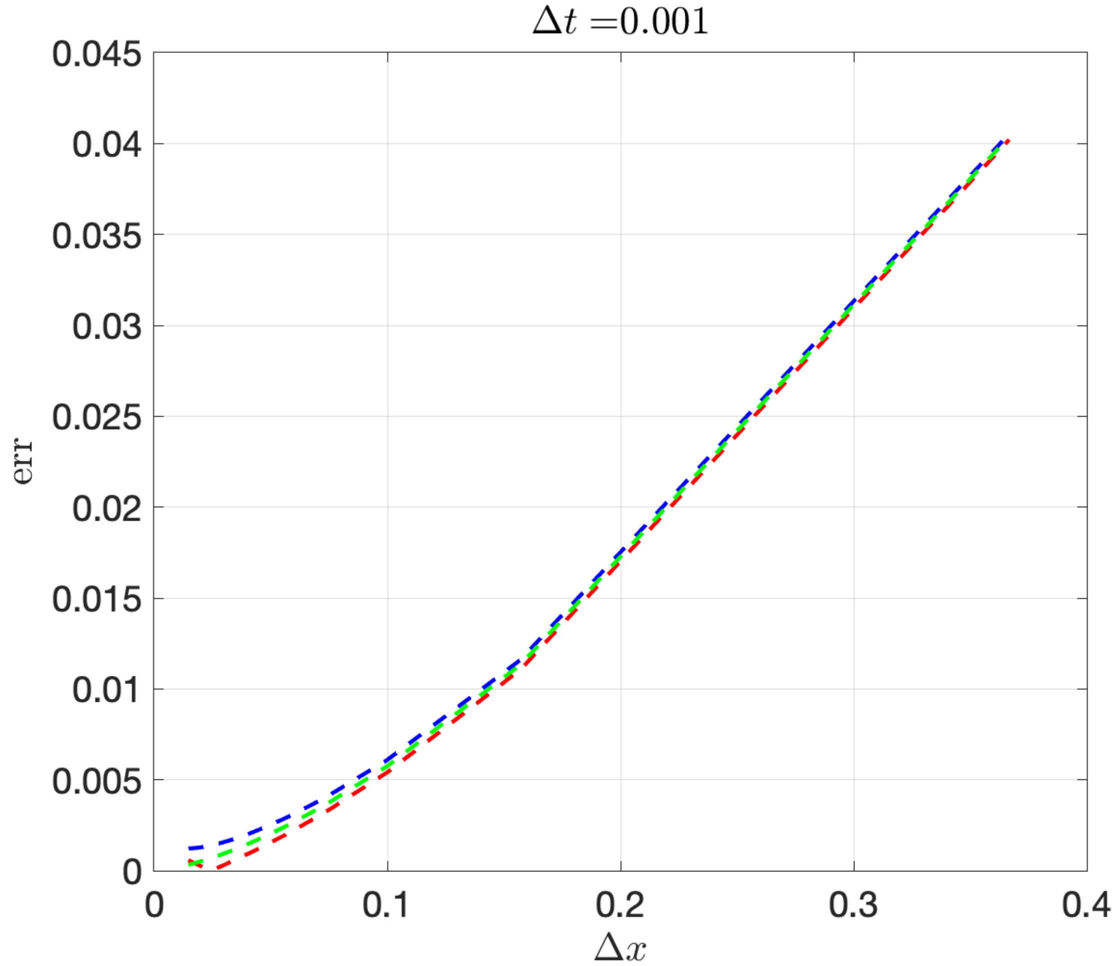


Figure 6 Similar but for varying delta-x. Only stable FTCS regimes are plotted.

5. NUMERICAL DISPERSION

5.1. Linear Advection

How different discretization methods impact the behavior of solutions. Start with the simplest examples. The linear advection in 1D:

$$\frac{\partial \theta}{\partial t} + c\frac{\partial \theta}{\partial x} = 0$$

We can use a solution in the form:

$$\theta(x,t) = \theta_0 e^{i(kx-\omega t)}$$

and the frequency of each mode satisfies a dispersion relation obtained by substituting the assumed solution into the governing equation:

$$\omega = ck$$

The results waves are non-dispersive given the phase speed $\left(c = {}^{\omega}/_{k}\right)$ and the group speed $\left({}^{\partial\omega}/_{\partial k} = c\right)$ are equal and constant. Let's see what the characteristics of the dispersion relation are by applying different discretization methods to the spatial derivative (assuming the temporal derivative is perfect).

Centered second order:

$$\frac{\partial \theta}{\partial t} + \frac{c}{2\Delta x}(\theta_{i+1} - \theta_{i-1}) = 0$$

Fourth order:

$$\frac{\partial \theta}{\partial t} + \frac{c}{12\Delta x}(\theta_{i-2} - 8\theta_{i-1} + 8\theta_{i+1} + \theta_{i+2}) = 0$$

Substituting in our solution of the form $e^{i(kx-\omega t)}$ after some manipulation, gives the following dispersion relations:

$$-i\omega = -\frac{c}{2\Delta x}\left(e^{ik\Delta x} - e^{-ik\Delta x}\right) \xrightarrow{yields} -i\omega = -\frac{ci}{\Delta x} \sin k\Delta x$$

and

$$-i\omega = -\frac{ci}{12\Delta x}\left(-e^{2ik\Delta x} + 8e^{ik\Delta x} - 8e^{-ik\Delta x} + e^{-2ik\Delta x}\right) \xrightarrow{yields} -i\omega$$
$$= -\frac{ci}{6\Delta x}(-sin2\,k\Delta x + 8sin\,k\Delta x)$$

for the second and fourth order respectively. The fourth order method is more accurate for a larger range of wave numbers but because the frequency then drops to zero over a shorted span of wave numbers the group speed error is larger. When plotted (Figure 7) we see that the frequencies are more accurate, however the grid-scale mode is stationary and so the group speed of the shortest waves is even worse than with the fourth order scheme. The pattern that emerges is that the grid-scale modes are always stationary and that increasing the order of the scheme does not necessarily improve the behaviour of the shortest scales.

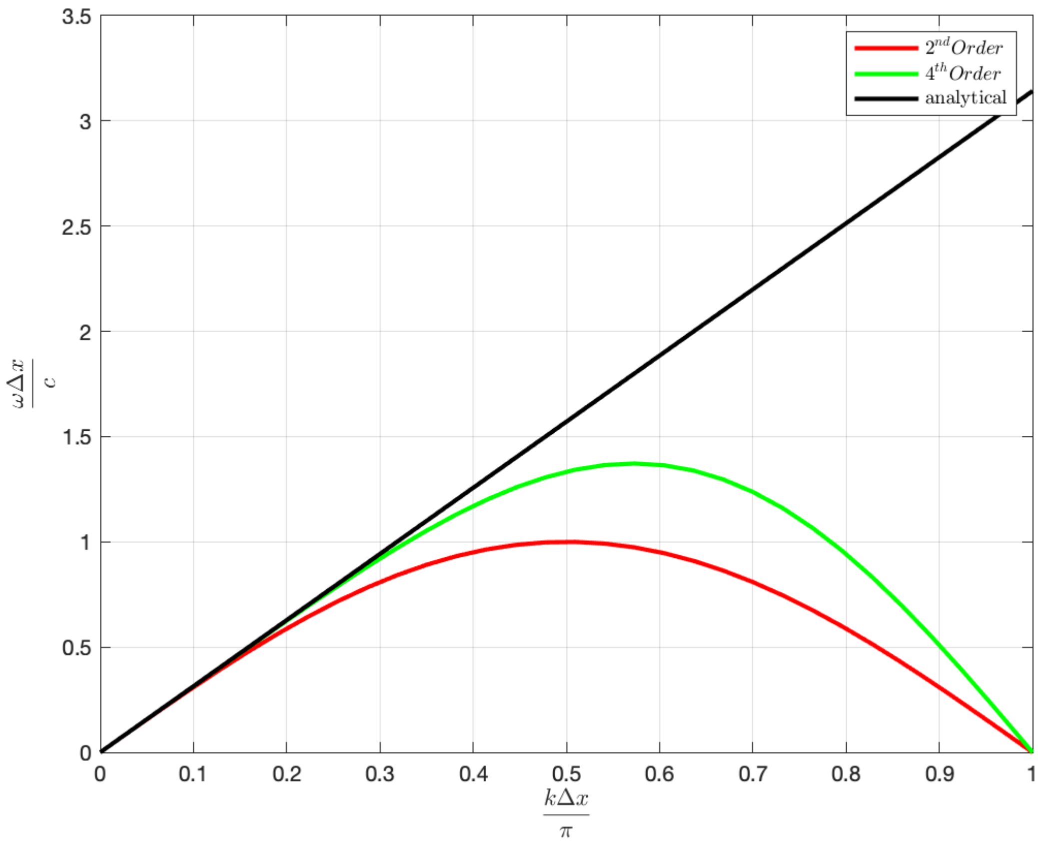


Figure 7 Dispersion relation for the linear advection equation. Second (red) and fourth (green) order accurate.

5.2. Gravity waves

Consider the adjustment of a non-rotating, constant depth shallow water layer. The governing linear equations in one dimension are:

$$\frac{\partial u}{\partial t} = -g\frac{\partial \eta}{\partial x}$$

$$\frac{\partial \eta}{\partial t} = -H\frac{\partial u}{\partial x}$$

where *g* is gravity, *H* is layer depth, *u* is the velocity and η the free-surface. We want to get a dispersion relation and then compare analytical solution with its numerical representation. Deriving the second by *t* and the first by *x* and substituting:

$$\frac{\partial^2 \eta}{\partial t^2} = -H\frac{\partial^2 u}{\partial t \partial x}$$

$$\frac{\partial^2 u}{\partial t \partial x} = -g\frac{\partial^2 \eta}{\partial x^2}$$

$$\frac{\partial^2 \eta}{\partial t^2} = Hg\frac{\partial^2 \eta}{\partial x^2}$$

Assuming a wave-like solution,

$$\eta = \eta_0 e^{i(kx-\omega t)}$$

And substituting, we obtain the dispersion relation:

$$\omega^2 = Hgk^2$$

which is quadratic and has two roots, so that waves can travel in both directions. The waves are non-dispersive and have group and phase speeds $c_g = \pm\sqrt{gH}$.

Now, consider the second order spatial discretization (still assuming perfect time derivative) and see what the characteristic of the resulting waves are. We proceed discretizing the spatial derivative in the surface elevation then substitute in the second equation and derive again (as a model would do):

$$\frac{\partial u}{\partial t} = -\frac{g}{2\Delta x}(\eta_{i+1} - \eta_{i-1})$$

$$\frac{\partial^2\eta}{\partial t^2} = -\frac{H}{2\Delta x}\left[-\frac{g}{2\Delta x}(\eta_{i+1}-\eta_{i-1})\Big|_{i+1} + \frac{g}{2\Delta x}(\eta_{i+1}-\eta_{i-1})\Big|_{i-1}\right]$$

$$\frac{\partial^2\eta}{\partial t^2} = -\frac{gH}{4\Delta x^2}[-(\eta_{i+1}-\eta_{i-1})|_{i+1} + (\eta_{i+1}-\eta_{i-1})|_{i-1}]$$

$$\frac{\partial^2\eta}{\partial t^2} = \frac{gH}{4\Delta x^2}(\eta_{i+2} - 2\eta_i+\eta_{i-2})$$

Substituting our assumed solution:

$$-\omega^2 = \frac{gH}{4\Delta x^2}\left(e^{i2k\Delta x} - 2 + e^{-i2k\Delta x}\right)$$

$$-\omega^2 = \frac{gH}{4\Delta x^2}(2\,cos(2k\Delta x) - 2)$$

The curves for the continuum and numerical solution are plotted in Fig. 9 (normalized wave number and freq.). From these curves we note the following:

- the numerical gravity waves are dispersive while the continuum is non-dispersive,
- a false maximum in the frequency occurs at $k = \frac{\pi}{2\Delta x}$ or $k' = \frac{1}{2}$
- the effective group speed has the wrong sign for wavelengths shorter than $4\Delta x$
- the Nyquist mode $\left(k = \frac{\pi}{\Delta x}\right)$ has zero frequency.

The last two points are symptomatic of the presence of a computational mode and these often occur when averaging operators are present. When the group speed has the wrong sign, energy is not radiated properly in the adjustment process and is often seen as "noise" near sources of energy.

We saw that evaluating derivative at middle point is four time more accurate than evaluating in the grid point. We can define a grid where different variables are "placed" in different points. Those are called staggered grids. Staggered differences become natural choices if we stagger the model variables in space (Fig. 8).

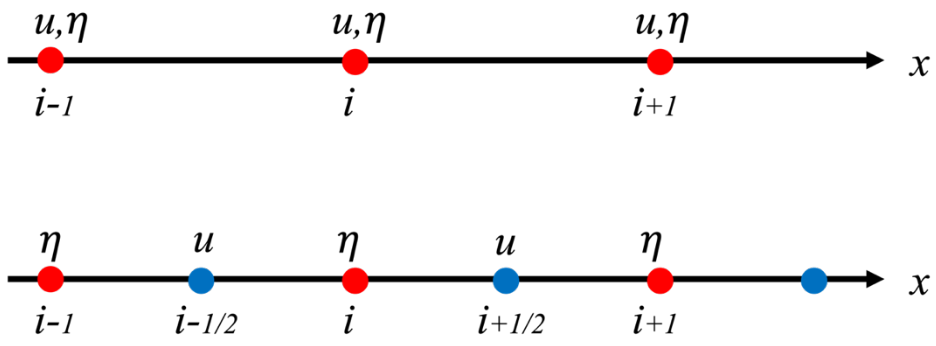


Figure 8 Example of staggered grid for the gravity waves

The discrete model then becomes:

$$\frac{\partial u}{\partial t} = -\frac{g}{\Delta x}\left(\eta_{i+1/2} - \eta_{i-1/2}\right)$$

$$\frac{\partial^2 \eta}{\partial t^2} = \frac{H}{\Delta x}\left[\frac{g}{\Delta x}\left(\eta_{i+1/2} - \eta_{i-1/2}\right)\Big|_{i+1/2} - \frac{g}{\Delta x}\left(\eta_{i+1/2} - \eta_{i-1/2}\right)\Big|_{i-1/2}\right]$$

$$\frac{\partial^2 \eta}{\partial t^2} = \frac{gH}{\Delta x^2}\left(\eta_{i+1} - 2\eta_i + \eta_{i-2}\right)$$

Substituting our assumed solution:

$$-\omega^2 = \frac{gH}{\Delta x^2}\left(2\cos(k\Delta x) - 2\right)$$

The dispersion relation is plotted in green in Fig. 9 and we note the following:

- when compared to the continuum we see that the numerical modes are still dispersive,
- there is no false maxima, unlike the non-staggered grid,
- the group speed is therefore of the correct sign everywhere, even if reduced.

The remarkable improvement in behaviour of the staggered grid over the non-staggered grid is due to the complete absence of operators with null-spaces (such as the averaging operator).

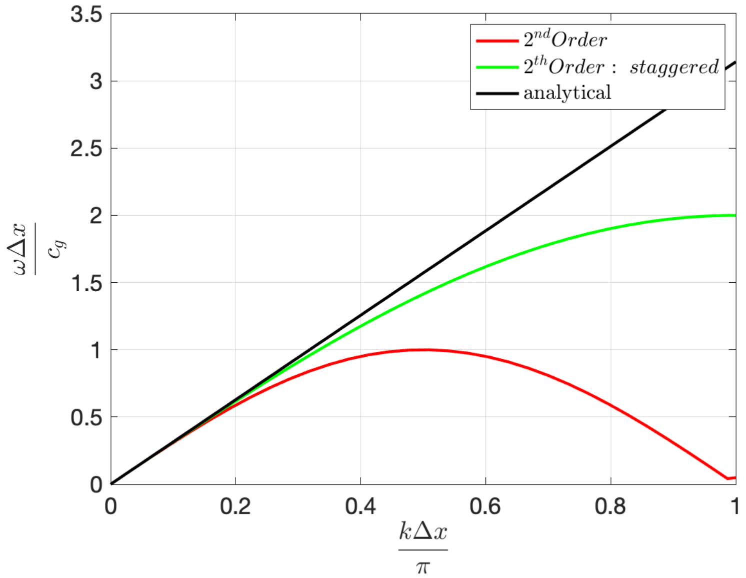


Figure 9 Dispersion relation for the gravity wave equation. Second centered (red) and second staggered (green) order accurate

5.3. Inertia-Gravity waves in 1D and 2D

5.3.1. 1D case

Consider now the geostrophic adjustment problem which introduces rotation. The relevant linear shallow water equations in one dimension are:

$$\frac{\partial u}{\partial t} - fv + g\frac{\partial \eta}{\partial x} = 0$$

$$\frac{\partial v}{\partial t} + fu = 0$$

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0$$

where gradients in the *y* direction have been dropped. This system adds one equation and one degree of freedom over the previous gravity wave problem. The dispersion relation can be obtained using a matrix method, representing the system by:

$$\begin{pmatrix} \frac{\partial}{\partial t} & -f & g\frac{\partial}{\partial x} \\ f & \frac{\partial}{\partial t} & 0 \\ H\frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial t} \end{pmatrix}\begin{pmatrix} u \\ v \\ \eta \end{pmatrix} = 0$$

Assuming solutions are in the form:

$$[u, v, \eta] = A_0 e^{i(kx-\omega t)}$$

and taking the determinant of the operator matrix

$$\left|\begin{pmatrix} -i\omega & -f & igk \\ f & -i\omega & 0 \\ iHk & 0 & -i\omega \end{pmatrix}\right| = 0 \Rightarrow \begin{cases} \omega = 0 \\ \omega^2 = f^2 + gHk^2 \end{cases}$$

To discretize the equations, we consider four possible configurations of variables on a regular grid, illustrated in Fig. 10 and labelled as A, B, C and D.

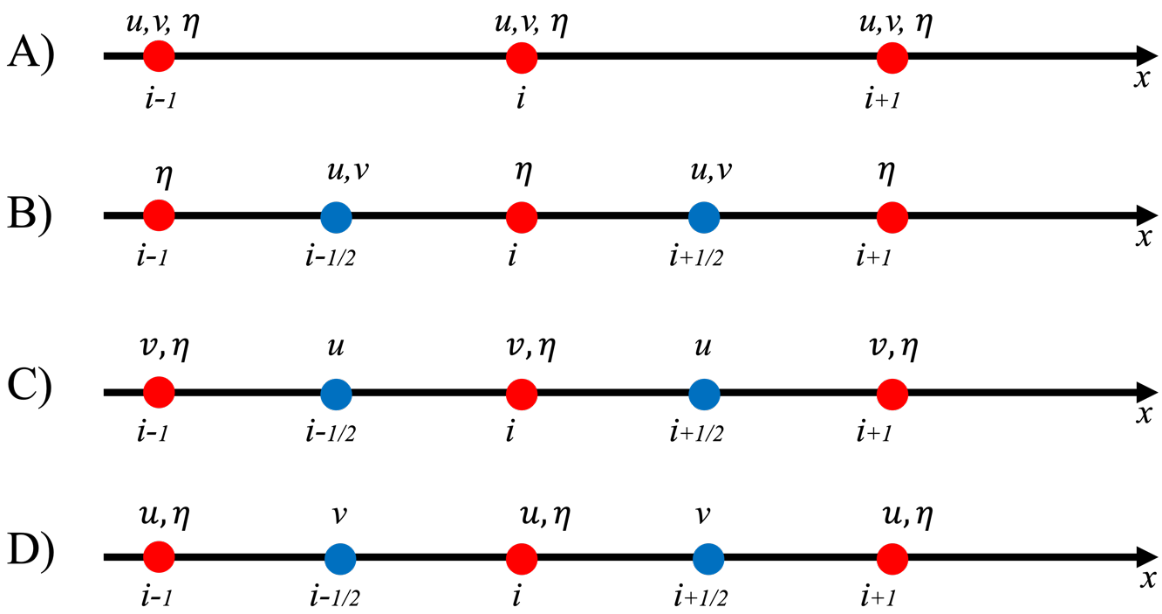


Figure 10 Different kind of grids

To simplify the treatment of the following, we introduce two basic operators, the center difference and the average or center linear interpolation. We define the operators as:

$$\delta_i f = f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}$$

$$\bar{f}^i = \frac{1}{2}\left(f_{i+\frac{1}{2}} + f_{i-\frac{1}{2}}\right)$$

So that the staggered center difference at the mid-point is:

$$f' \approx \frac{f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}}{\Delta x} = \frac{1}{\Delta x}\delta_i f$$

While the staggered evaluated at grid-point ($2\Delta x$):

$$f' \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x} = \frac{1}{\Delta x}\delta_i \bar{f}^i$$

The four discrete models based on each grid are then:

$$A\begin{cases} \dfrac{\partial u}{\partial t} - fv + \dfrac{g}{\Delta x}\delta_i \bar{\eta}^i = 0 \\ \dfrac{\partial v}{\partial t} + fu = 0 \\ \dfrac{\partial \eta}{\partial t} + \dfrac{H}{\Delta x}\delta_i \bar{u}^i = 0 \end{cases} \qquad B\begin{cases} \dfrac{\partial u}{\partial t} - fv + \dfrac{g}{\Delta x}\delta_i \eta = 0 \\ \dfrac{\partial v}{\partial t} + fu = 0 \\ \dfrac{\partial \eta}{\partial t} + \dfrac{H}{\Delta x}\delta_i u = 0 \end{cases}$$

$$C\begin{cases} \dfrac{\partial u}{\partial t} - f\bar{v}^i + \dfrac{g}{\Delta x}\delta_i \eta = 0 \\ \dfrac{\partial v}{\partial t} + f\bar{u}^i = 0 \\ \dfrac{\partial \eta}{\partial t} + \dfrac{H}{\Delta x}\delta_i u = 0 \end{cases} \qquad D\begin{cases} \dfrac{\partial u}{\partial t} - f\bar{v}^i + \dfrac{g}{\Delta x}\delta_i \bar{\eta}^i = 0 \\ \dfrac{\partial v}{\partial t} + f\bar{u}^i = 0 \\ \dfrac{\partial \eta}{\partial t} + \dfrac{H}{\Delta x}\delta_i \bar{u}^i = 0 \end{cases}$$

And the corresponding dispersion relations:

$$A\left\{\frac{\omega^2}{f^2} = 1 + \frac{4L_d^2}{\Delta x^2} s_k^2 c_k^2\right.$$

$$B\left\{\frac{\omega^2}{f^2} = 1 + \frac{4L_d^2}{\Delta x^2} s_k^2\right.$$

$$C\left\{\frac{\omega^2}{f^2} = c_k^2 + \frac{4L_d^2}{\Delta x^2} s_k^2\right.$$

$$D\left\{\frac{\omega^2}{f^2} = c_k^2 + \frac{4L_d^2}{\Delta x^2} s_k^2 c_k^2\right.$$

where $L_d = \sqrt{gH}/f$ is the radius of deformation and $s_k = sin(k\Delta x/2)$ and $c_k = \cos(k\Delta x/2)$ are common factors. The coefficient in front of the gravity-wave contribution is a resolution parameter:

$$R^2 = \frac{L_d^2}{\Delta x^2}$$

and it controls the form of the some of the dispersion relations. The Rossby radius of deformation is the scale distance over which the gravitational tendency to render the free surface flat is balanced by the tendency of Coriolis acceleration to deform the surface.

Let's find the *A*-grid solution step-by-step, *B*-grid solution is straightforward, and then the student should find the other grids solution. The operator $\delta_i \overline{f'}$ in case of a solution of the form $e^{i(kx-\omega t)}$ is:

$$\delta_i \overline{f'} = e^{i(k(n+1)\Delta x-\omega t)} - e^{i(k(n-1)\Delta x-\omega t)} = e^{i(kn\Delta x-\omega t)} \cdot e^{i(k\Delta x)} - e^{i(kn\Delta x-\omega t)} \cdot e^{-i(k\Delta x)}$$
$$= e^{i(kn\Delta x-\omega t)}\left(e^{i(k\Delta x)} - e^{-i(k\Delta x)}\right) = e^{i(kn\Delta x-\omega t)}\, isin(k\Delta x)$$

If we omit the term $e^{i(kn\Delta x-\omega t)}$, the operator matrix form is:

$$\left|\begin{pmatrix} -i\omega & -f & i\dfrac{g}{\Delta x}\sin(k\Delta x) \\ f & -i\omega & 0 \\ i\dfrac{H}{\Delta x}\sin(k\Delta x) & 0 & -i\omega \end{pmatrix}\right|$$

forcing its determinant to zero and omitting the zero-frequency solution we get:

$$\frac{\omega^2}{f^2} = 1 + \frac{gH}{f^2\Delta x^2}\sin^2(k\Delta x)$$

Substituting the radius definition, L_d, and knowing that:

$$\sin^2(k\Delta x) = 4\sin^2\left(\frac{k\Delta x}{2}\right)\cos^2\left(\frac{k\Delta x}{2}\right)$$

We get:

$$\frac{\omega^2}{f^2} = 1 + \frac{4L_d^2}{\Delta x^2}s_k^2 c_k^2$$

The dispersion relation of the continuous and four numerical inertia-gravity waves are plotted in Fig. 11. In panel (a) the resolution parameter is larger than one meaning the deformation radius is resolved. In panel (b) the resolution parameter is less than one meaning the deformation radius is not resolved.

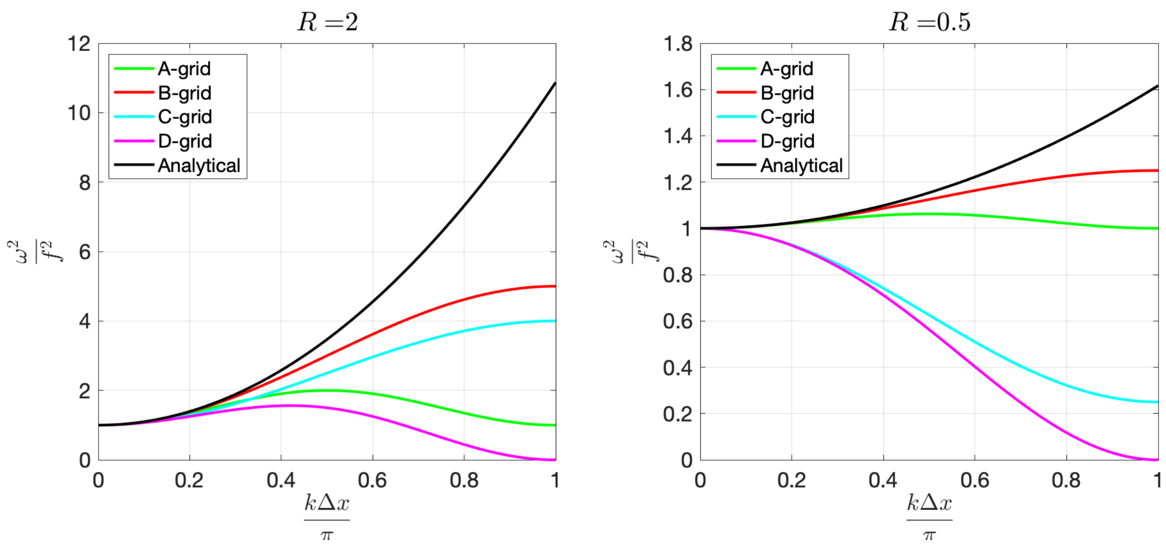


Figure 11 The dispersion relation of the continuous and four numerical inertia-gravity waves

In the continuum, the frequency increases monotonically with wave number, the waves are dispersive and the group and phase speeds are pointed in the direction of the wave vector. Instead in the numerical solution we observe:

- Scheme *A*, regardless of resolution has a false maximum at $k'=1/2$ so that grid-scale waves ($k'=1$) oscillate with the inertial frequency rather than propagate as gravity waves,
- Scheme *D* has false maxima and it is worse in that the grid-scale waves become stationary,
- Scheme *C* has the wrong slope (group speed in wrong direction) if $R < 1$ but is qualitatively correct if $R > 1$,
- Scheme *B* is qualitatively correct for both values of *R* and so appears to be the best scheme.

5.3.2. 2D case

Now we extend our liner system to the 2D case. We consider only grids type-*A*, *B* and *C* as shown in Fig. 12.

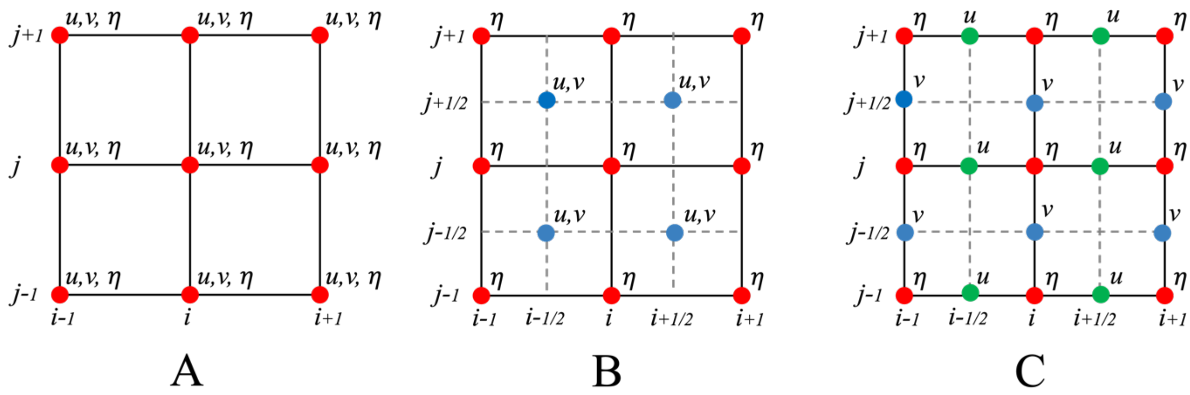


Figure 12 2D horizontal grids

The set of governing equations is now:

$$\frac{\partial u}{\partial t} - fv = -g\frac{\partial \eta}{\partial x}$$

$$\frac{\partial v}{\partial t} + fu = -g\frac{\partial \eta}{\partial y}$$

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0$$

We seek solution of the form:

$\eta = \eta_0 e^{i(kx+ly-\omega t)}$, $u = u_0 e^{i(kx+ly-\omega t)}$, $v = v_0 e^{i(kx+ly-\omega t)}$

The dispersion relation can be obtained again using a matrix method, representing the system by:

$$\begin{pmatrix} \frac{\partial}{\partial t} & -f & g\frac{\partial}{\partial x} \\ f & \frac{\partial}{\partial t} & g\frac{\partial}{\partial y} \\ H\frac{\partial}{\partial x} & H\frac{\partial}{\partial y} & \frac{\partial}{\partial t} \end{pmatrix}\begin{pmatrix} u \\ v \\ \eta \end{pmatrix} = 0$$

and taking the determinant of the operator matrix

$$\left|\begin{pmatrix} -i\omega & -f & igk \\ f & -i\omega & igl \\ iHk & iHl & -i\omega \end{pmatrix}\right| = 0 \Rightarrow \begin{cases} \omega = 0 \\ \omega^2 = f^2 + gH(k^2 + l^2) \end{cases}$$

The three discrete models are then:

$$A\begin{cases}\dfrac{\partial u}{\partial t}-fv+\dfrac{g}{\Delta x}\delta_i\bar{\eta}^i=0\\ \dfrac{\partial v}{\partial t}+fu+\dfrac{g}{\Delta y}\delta_j\bar{\eta}^j=0\\ \dfrac{\partial \eta}{\partial t}+\dfrac{H}{\Delta x}\delta_i\bar{u}^i+\dfrac{H}{\Delta y}\delta_j\bar{v}^j=0\end{cases}\qquad B\begin{cases}\dfrac{\partial u}{\partial t}-fv+\dfrac{g}{\Delta x}\delta_i\bar{\eta}^j=0\\ \dfrac{\partial v}{\partial t}+fu+\dfrac{g}{\Delta y}\delta_j\bar{\eta}^i=0\\ \dfrac{\partial \eta}{\partial t}+\dfrac{H}{\Delta x}\delta_i\bar{u}^j+\dfrac{H}{\Delta y}\delta_j\bar{v}^i=0\end{cases}$$

$$C\begin{cases}\dfrac{\partial u}{\partial t}-f\bar{v}^{ij}+\dfrac{g}{\Delta x}\delta_i\eta=0\\ \dfrac{\partial v}{\partial t}+f\bar{u}^{ij}+\dfrac{g}{\Delta y}\delta_j\eta=0\\ \dfrac{\partial \eta}{\partial t}+\dfrac{H}{\Delta x}\delta_i u+\dfrac{H}{\Delta y}\delta_j v=0\end{cases}$$

Let derive the grid-A solutions. The discrete operator matrix is:

$$\left\|\begin{pmatrix}-i\omega & -f & i\dfrac{g}{\Delta x}\sin(k\Delta x)\\ f & -i\omega & i\dfrac{g}{\Delta y}\sin(l\Delta y)\\ i\dfrac{H}{\Delta x}\sin(k\Delta x) & i\dfrac{H}{\Delta y}sin(l\Delta y) & -i\omega\end{pmatrix}\right\|$$

forcing its determinant to zero, omitting the zero-frequency solution:

$$\omega^2=f^2+\frac{gH}{\Delta x^2}[sin^2(k\Delta x)]+\frac{gH}{\Delta y^2}[sin^2(l\Delta y)]$$

Defining, similar to the one dimension, $s_l=sin(l\Delta y/2)$ and $c_l=\cos(l\Delta y/2)$ and assuming $\Delta x\equiv\Delta y$ we can write:

$$A\left\{\omega^2=f^2+\frac{4gH}{\Delta x^2}[s_k^2c_k^2+s_l^2c_l^2]\right.$$

Similarly for the *B* and *C* grids the dispersion relations are:

$$B\left\{\omega^2=f^2+\frac{4gH}{\Delta x^2}[s_k^2c_l^2+s_l^2c_k^2]\right.$$

$$C\left\{\omega^2=f^2c_k^2c_l^2+\frac{4gH}{\Delta x^2}[s_k^2+s_l^2]\right.$$

We anticipate that to obtain the satisfactory result, great care must be taken on the value of *R*. Shown in Figure 13 are comparison of the frequencies by using two values of R. The left column shows the frequency with R = 2 while the right column shows the frequency with R = 0.25. As can be seen from the frequency plots for $R = 2$, the continuum frequency increases monotonically with wavenumber. The frequencies for both *A* and *B* grid agree with the continuum frequency only in the third quadrant. Outside third quadrant, the frequency for A and B grid decreases as the wavenumber increases which contradict with the continuum frequency. Unlike the other frequencies, the frequency for C grid shows the same monotonic increase of frequency as the continuum frequency. For $R=0.25$, both frequencies for *A* and *B* grid show the same distortion as $R = 2$. However, the C-grid frequency now shows a monotonic decrease with increasing wavenumber. Hence all discrete frequencies contrary to the continuum frequency. Thus, for the two-dimensional case with rotation *C*-grid is the most satisfactory choice of grid in producing a better solution with an appropriate value of *R*.

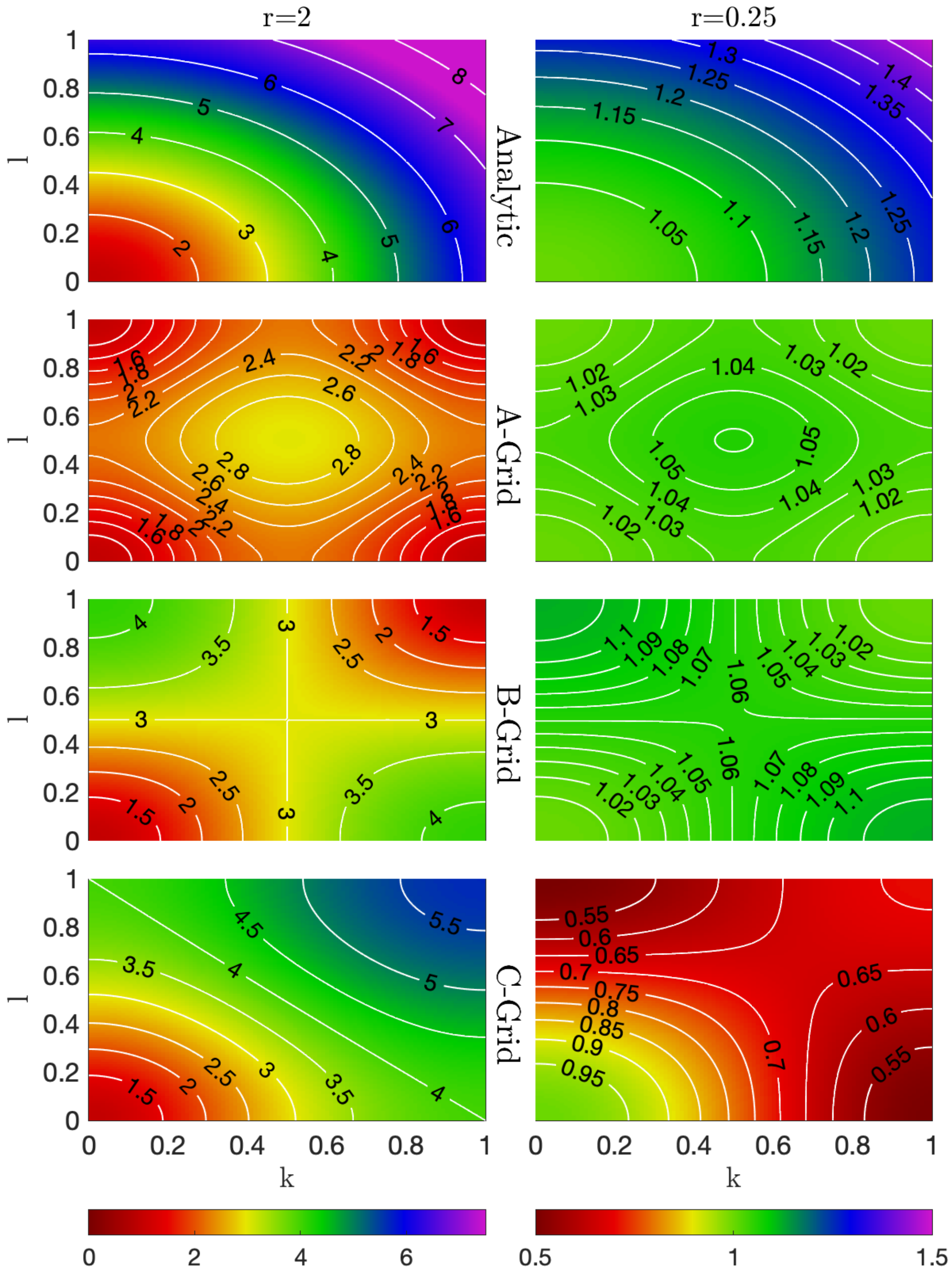


Figure 13 dispersion relation for inertia-gravity in 2D case. Only for analytical solution, Grid-A, Grid-B and Grid-C.

6. NUMERICAL DIFFUSIVITY (Modified equations)

Our tool for estimating the accuracy of a finite difference method has been the local truncation error. A slightly different approach based on "modified equation" can be very illuminating revealing much more about the structure and behaviour of the numerical solution. Modified equations provide a way of anticipating the behaviour of a numerical method by examining terms implied by a particular approximation. For a given difference equation, the corresponding modified equation is a continuous equation for which the same difference equation is a higher order approximation. If we can find such equation, then studying the behaviour of solutions to this PDE should tell us much about how the numerical approximation is behaving. This can be advantageous because it is often easier to study the behaviour of PDE's than of finite difference formulas.

If we consider a simple linear advection equation:

$$\frac{\partial \phi}{\partial t} + c\frac{\partial \phi}{\partial x} = 0$$

Now assume to find the discretized solution using a Forward in Time Up-wind in Space (FTUS) method. The upwind scheme uses a side-difference in space biased in the upwind direction so that for positive flow (c>0) the scheme is:

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} + \frac{c}{\Delta x}(\phi_i^m - \phi_{i-1}^m) = 0$$

The truncation term of the time derivative is:

$$\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t} = \frac{\partial \phi}{\partial t} + \frac{\Delta t}{2!}\frac{\partial \phi^2}{\partial t^2} + \frac{\Delta t^2}{3!}\frac{\partial \phi^3}{\partial t^3} + \ldots$$

That is, while $\frac{\phi_i^{m+1} - \phi_i^m}{\Delta t}$ is first order accurate for $\frac{\partial \phi}{\partial t}$, it is second order accurate for $\frac{\partial \phi}{\partial t} + \frac{\Delta t}{2}\frac{\partial \phi^2}{\partial t^2}$. Similarly for the space derivative:

$$\frac{c}{\Delta x}(\phi_i^m - \phi_{i-1}^m) = c\frac{\partial \phi}{\partial x} - \frac{\Delta x}{2!}\frac{\partial \phi^2}{\partial x^2} + \frac{\Delta x^2}{3!}\frac{\partial \phi^3}{\partial x^3} + \ldots$$

Is first order accurate for $c\frac{\partial\phi}{\partial x}$ while its is second order accurate for $c\frac{\partial\phi}{\partial x} - \frac{c\Delta x}{2}\frac{\partial\phi^2}{\partial x^2}$. This is an interesting result. Our FTUS is first order accurate for the advection equation but is second order accurate in representing:

$$\frac{\partial\phi}{\partial t} + c\frac{\partial\phi}{\partial x} + \frac{1}{2}\left(\Delta t\frac{\partial\phi^2}{\partial t^2} - c\Delta x\frac{\partial\phi^2}{\partial x^2}\right) = 0$$

To better see the physics represented in this equation we can differentiate with respect to time:

$$\frac{\partial\phi^2}{\partial t^2} + c\frac{\partial\phi^2}{\partial t\partial x} + \frac{1}{2}\left(\Delta t\frac{\partial\phi^3}{\partial t^3} - c\Delta x\frac{\partial\phi^3}{\partial t\partial x^2}\right) = 0$$

space:

$$\frac{\partial\phi^2}{\partial t\partial x} + c\frac{\partial\phi^2}{\partial x^2} + \frac{1}{2}\left(\Delta t\frac{\partial\phi^3}{\partial x\partial t^2} - c\Delta x\frac{\partial\phi^3}{\partial x^3}\right) = 0$$

Substitute and omit term of first order in Δt or Δx:

$$\frac{\partial\phi^2}{\partial t^2} = c^2\frac{\partial\phi^2}{\partial x^2} + O(\Delta t, \Delta x)$$

We can now insert the second time derivative in our equation to get:

$$\frac{\partial\phi}{\partial t} + c\frac{\partial\phi}{\partial x} - \frac{c\Delta x}{2}\left(1 - \frac{c\Delta t}{\Delta x}\right)\frac{\partial\phi^2}{\partial x^2} = 0$$

This is now a familiar advection-diffusion equation. The modified equation differs from the governing equation by a diffusive term with diffusivity:

$$\frac{c\Delta x}{2}\left(1 - \frac{c\Delta t}{\Delta x}\right)$$

This numerical diffusivity is proportional to the flow c, it decreases with decreasing Δx (i.e. higher resolution is less diffusive) and also decreases with the Courant number $\frac{c\Delta t}{\Delta x}$. The fact that the modified equation is an advection-diffusion equation tells us a great deal about how the numerical solution behaves. Note that the diffusion coefficient vanishes in the special case. Also note that the diffusion coefficient is positive only if $0 < \frac{c\Delta t}{\Delta x} < 1$. This is precisely the stability limit of upwind. If this is violated, then the diffusion coefficient in the modified equation is negative, giving an ill-posed problem with exponentially growing solutions. Hence, we see that even some information about stability can be extracted from the modified equation.

7. MORE ON THE ADVECTION SCHEMES

We have seen the up-wind, and its characteristics. In general, sided algorithms of odd order tend to be diffusive. Alternative methods were proposed in 1960 by P.D. Lax and B. Wendroff for solving systems of hyperbolic conservation laws on the generic form

$$\frac{\partial \phi}{\partial t} + \frac{\partial c\phi}{\partial x} = 0$$

So that

$$\frac{\partial \phi}{\partial t} + \frac{\partial F}{\partial x} = 0$$

Where F is now the advective flux $F(u)=c\phi$, and this is called the flux form of the advection equation. A large class of numerical methods for solving flux form of the advection equation are the called conservative methods:

$$\phi_i^{m+1} = \phi_i^m + \frac{\Delta t}{\Delta x}\left(F_{i-\frac{1}{2}} - F_{i+\frac{1}{2}}\right)$$

The Lax–Wendroff method can be derived in various ways. For simplicity, we will derive the method by using a simple linear advection equation with $F(u)=c\phi$, where c is a constant propagation velocity. We start with a Taylor approximation of ϕ_i^{m+1}:

$$\phi_i^{m+1} = \phi_i^m + \Delta t\frac{\partial \phi}{\partial t} + \frac{\Delta t^2}{2}\frac{\partial \phi^2}{\partial t^2} + \ldots$$

Knowing that:

$$\frac{\partial \phi}{\partial t} = -c\frac{\partial \phi}{\partial x}$$

and

$$\frac{\partial \phi^2}{\partial t^2} = c^2\frac{\partial \phi^2}{\partial x^2} + O(\Delta t, \Delta x)$$

We can write:

$$\phi_i^{m+1} = \phi_i^m - c\Delta t\frac{\partial \phi}{\partial x} + c^2\frac{\Delta t^2}{2}\frac{\partial \phi^2}{\partial x^2} + \ldots$$

Now discretizing the spatial derivatives as:

$$\frac{\partial \phi}{\partial x} \approx \frac{\phi_{i+1}^{m} - \phi_{i-1}^{m}}{2\Delta x}$$

$$\frac{\partial \phi^2}{\partial x^2} \approx \frac{\phi_{i+1}^{m} - 2\phi_{i}^{m} + \phi_{i-1}^{m}}{\Delta x^2}$$

Combining we get:

$$\phi_{i}^{m+1} = \phi_{i}^{m} - \frac{c\Delta t}{2\Delta x}(\phi_{i+1}^{m} - \phi_{i-1}^{m}) + \frac{c^2 \Delta t^2}{2\Delta x^2}(\phi_{i+1}^{m} - 2\phi_{i}^{m} + \phi_{i-1}^{m})$$

The Lax-Wendroff method is basically a FTCS scheme plus a diffusion term. The diffusion term cancels the leading truncation error from the forward time difference, the method is in fact second order accurate in time and space. This is an example of how treating time and space together leads to a substantially different scheme than would be obtained by discretizing the dimensions independently.

The FTUS and Lax-Wendroff methods each have their advantages and disadvantages; the FTUS is first order accurate and very diffusive but does conserve extrema while the Lax-Wendroff scheme doesn't conserve extrema but is second order accurate. Using a first order scheme require a very high resolution (and/or a cfl number close to one) in order to get a satisfying solution. On the other hand, if the solution has big gradients or discontinuities, the second order schemes fail, and introduce nonphysical oscillations due to their dispersive nature. That is, the second order schemes give a much higher accuracy on smooth solutions, than the first order schemes, whereas the first order schemes behave better in regions containing sharp gradients or discontinuities.

The basic idea of the flux limiter scheme is to combine the two methods capturing the desired features of each.

We start writing the system in the Flux form:

$$\phi_{i}^{m+1} = \phi_{i}^{m} + \frac{\Delta t}{\Delta x}\left(F_{i-\frac{1}{2}} - F_{i+\frac{1}{2}}\right)$$

And we define the total advective fluxes as the combination of the two individual fluxes:

$$F_{i-\frac{1}{2}} = F^{UW}_{i-\frac{1}{2}} + \psi(r)\left(F^{LW}_{i-\frac{1}{2}} - F^{UW}_{i-\frac{1}{2}}\right)$$

$$F_{i+\frac{1}{2}} = F^{UW}_{i+\frac{1}{2}} + \psi(r)\left(F^{LW}_{i+\frac{1}{2}} - F^{UW}_{i+\frac{1}{2}}\right)$$

where $\psi(r)$ is the limiter function, r is a measure of the smoothness of the solution, F^{LW} and F^{UW} are the fluxes computed according the Lax–Wendroff or the Up-Wind method respectively. In general, we can substitute the up-wind and the Lax-Wendroff methods with any other first and high order schemes. The limiter function ψ is designed to be equal to 1 (one) in regions where the solution is smooth, in which F reduces to F^{LW}, a pure high order scheme:

$$F^{UW}_{i+\frac{1}{2}} = c\phi_i^m$$

$$F^{LW}_{i+\frac{1}{2}} = c\phi_i^m + \frac{c(1-C)}{2}(\phi_{i+1}^m - \phi_i^m)$$

with $C = \frac{c\Delta t}{\Delta x}$ the Courant number.

In regions where the solution is not smooth (sharp gradients or discontinuities) ψ is designed to be equal to zero, in which F reduces to F^{UW}, and a pure low order scheme. As a measure of the smoothness, r is commonly taken to be the ratio of consecutive gradients:

$$r_{i-\frac{1}{2}} = \frac{\phi_{i-1}^m - \phi_{i-2}^m}{\phi_i^m - \phi_{i-1}^m}$$

$$r_{i+\frac{1}{2}} = \frac{\phi_i^m - \phi_{i-1}^m}{\phi_{i+1}^m - \phi_i^m}$$

So that our discretized linear advection equation can be written as:

$$\phi_i^{m+1} = \phi_i^m - C\left[1 - \frac{1-C}{2}\psi_{i-\frac{1}{2}} + 1 - \frac{1-C}{2}\frac{\psi_{i+\frac{1}{2}}}{r_{i+\frac{1}{2}}}\right](\phi_i^m - \phi_{i-1}^m)$$

which looks like an upwind scheme with a more complex Courant number. The upwind scheme is both monotone and stable if the “effective” Courant number is both non-negative and less than one:

$$0 \leq -C\left[1 - \frac{1-C}{2}\psi_{i-\frac{1}{2}} + 1 - \frac{1-C}{2}\frac{\psi_{i+\frac{1}{2}}}{r_{i+\frac{1}{2}}}\right] \leq 1$$

Given $C \geq 0$ the above condition is surely satisfied if:

$$\left|\frac{\psi_{i+\frac{1}{2}}}{r_{i+\frac{1}{2}}} - \psi_{i-\frac{1}{2}}\right| \leq 2$$

If $r < 0$, the slope must have changed sign and indicates a local extrema. In this instance we should limit the advective flux to take the form of the upwind flux since it is the only (linear) scheme capable of conserving extrema. Therefore, for r<0 we set ψ (r) = 0. If *r>0*, the above inequality is satisfied when

$$0 \leq \frac{\psi(r)}{r} \leq 2 \quad and\ 0 \leq \psi(r) \leq 2$$

and we must find functions that meet these criteria if the new scheme is to conserve extrema. It happens that the FTUS scheme is TVD.

There are many possible limiters but the most widely used are:

Superbee: $\psi(r)$ = *max(0,min(1,2r),min(2,r))*

minmod: $\psi(r)$= *max(0,min(1,r))*

Van Leer: $\psi(r) = \frac{r+|r|}{1+|r|}$

A flux limiter has the following effect:

- For smooth parts of a solution it will do second order accurate flux-conserved advection.
- For regions near a jump or a very sharp and sudden gradient it will switch to first order upwind flux-conserved advection.

Flux limiters therefore make sure that one gets the best of both second order and first order methods.

In Fig.14 the results from different schemes are shown together with the analytical solution. It is evident that the up-wind scheme is very diffusive and for longer integration (bottom panel) loses the structure of the wave. The high order accuracy generates spurious local minima and maxim. Different flux limiter implementations seem to be a good choice.

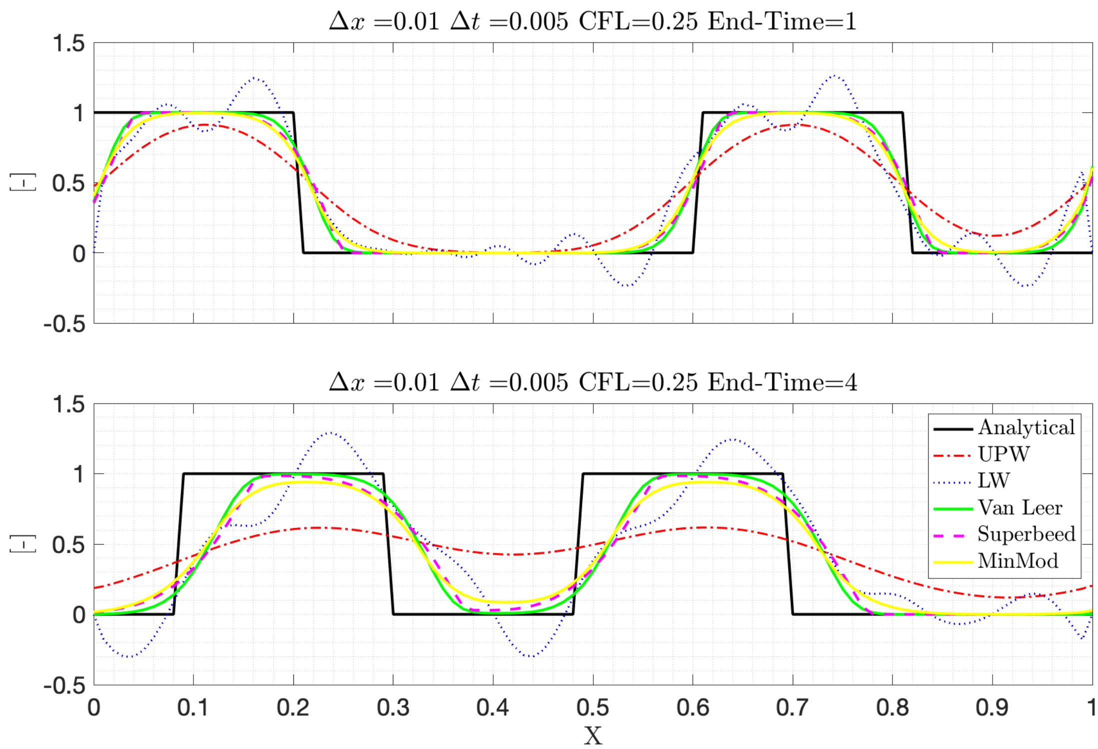


Figure 14

8. SUB-GRID SCALE PARAMETERIZATION

The need to solve our equations of motion on finite-resolution spatial and temporal grids implies the existence of "subgridscale" (SGS) processes (molecular diffusion and viscosity, three-dimensional turbulence, internal wave breaking, convection ...). The aggregate consequences of these SGS processes need to be included via parameterization of their interactions with dynamics explicitly resolved.

8.1. The Closure Problem

The earliest attempt to include some of the effects of smaller-scale processes on the larger scale dates to Reynolds (1895). The equations for the mean flow have been derived splitting the variable and then taking the average of the resulting equation. Solution of the mean equations calls for knowledge of the Reynolds stress tensor as a function of space and time. Since these terms are unknown *a priori* and cannot be represented in their own closed set of equations, we cannot solve for the mean quantities unless we somehow specify the turbulent stresses directly or specify their relationship to the mean fields. To obtain the equation for the fluctuating part we can subtract the equation for the mean component from the total component. Continuing we can construct kinetic energy equations for the mean and eddy contributions by multiplication with the respective velocity components and combining the two we can derive an equation for the mean turbulent kinetic equation:

$$k = \frac{1}{2}\langle {u_i}'{u_i}'\rangle = \frac{1}{2}(u'u', v'v', w'w')$$

$$\frac{\partial k}{\partial t} + \langle u_j\rangle\frac{\partial k}{\partial x_j} = -\frac{1}{\rho_0}\frac{\partial\langle {u_i}'p'\rangle}{\partial x_i} - \frac{1}{2}\frac{\partial\langle {u_j}'{u_j}'{u_i}'\rangle}{\partial x_i} + \nu\frac{\partial^2 k}{\partial x_j^2} - \langle {u_i}'{u_j}'\rangle\frac{\partial\langle {u_i}'\rangle}{\partial x_j} - \nu\,\langle\frac{\partial {u_i}'}{\partial x_j}\frac{\partial {u_i}'}{\partial x_j}\rangle$$
$$-\frac{g}{\rho_0}\langle\rho'{u_i}'\rangle\delta_{i3}$$

In these equations:

$$\langle {u_i}'{u_j}'\rangle\frac{\partial\langle {u_i}'\rangle}{\partial x_j}$$

represents a source term due to current shear and:

$$-\frac{g}{\rho_0}\langle\rho' u_i'\rangle\delta_{i3}$$

due to buoyancy. Note that a triple product between the turbulent variables appears in the equation for turbulent kinetic energy. Thus, new variables (that is, variables for which there are no prognostic equations) arise at each level in the Reynolds averaging procedure. Evolutionary equations for the triple correlation terms in the preceding TKE equation can, for instance, be derived. However, they involve further (fourth order) unknowns and so on for higher moment equations. This is the so-called closure problem.

8.2. Overview of SGS Closures

In ocean modelling, several different closure approaches are used. A first-order closure parameterizes the second-order moments (the Reynolds stresses) directly. Higher-order closure schemes use a prognostic equation for the variance and/or covariances, then parameterize the higher moments. A schematic "family tree" of subgridscale (SGS) mixing schemes is shown in Fig 15. Very important, the approaches for lateral and vertical SGS closure vary considerably. Except for simple first-order closures based upon mixing length arguments, strategies for horizontal and vertical mixing have evolved separately, in large part because the spatial scales and processes involved in each are quite different. Further distinctions among SGS schemes can be made based on the degree of physical motivation: filters and higher-order operators are often numerically necessary to remove small-scale noise but are largely ad-hoc in form and motivation, while special dynamical parameterizations are available for certain processes (topographic stress, eddy thickness diffusion, and convection).

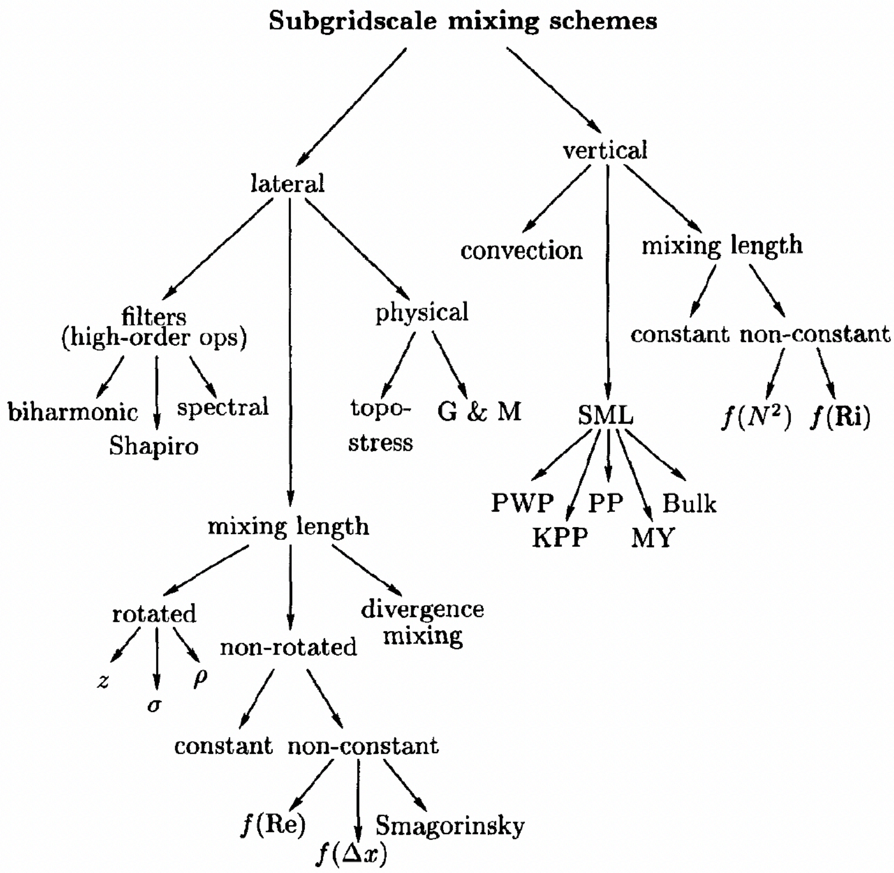


Figure 15 from Haidvogel and Beckmann, Numerical Ocean Circulation Modeling

In the vertical, the surface mixed layer (SML) has historically received special attention because of its important role in air-sea exchange. Adaptive (non-constant) mixing length schemes are widely used for parameterization of both lateral and vertical mixing. In the horizontal, parameterizations dependent on the rates of stress and strain (Smagorinsky), grid spacing, and Reynolds number have been advocated. In the vertical, mixing as a function of stability frequency and/or Richardson number are historically prevalent. Since mixing in the ocean does depend in some complex way on these (and other) parameters, these adaptive schemes may be viewed as simplified attempts to incorporate actual physical dependencies. An issue of some importance is the orientation of the mixing operators, particularly in the oceanic interior within which diapycnal mixing rates are weak. For certain classes of numerical models, weak diapycnal mixing may be difficult to achieve, and may depend on the rotation of the mixing tensors away from the native "lateral" coordinate directions. While there is no need for isopycnic models to rotate their along-coordinate mixing operator, this is not the case for geopotential and terrain-following coordinate models. Horizontal and along-sigma mixing has long been the default for these classes of models; however, algorithms for stable and conservative isopycnal rotation of diffusion tensors are now common. Often, viscous and diffusive operators are chosen to have identical forms, though perhaps with different dissipative coefficients. However, there may be reason to desire different properties of the SGS parameterization scheme for turbulent mixing of momentum and tracers. For example, the demands for an appropriate mixing scheme for tracers are more stringent than for momentum. Monotonicity and positive definiteness are usually considered important properties of the numerical solution for a tracer equation, but less so for the momentum equations. The choice of diffusive closure is therefore closely allied with the form the horizontal advection operator.

8.3. First Order Closures

8.3.1. Constant Eddy coefficients

The simplest approach is to parameterize the Reynolds stresses in terms of the largescale flow by assuming a linear dependence on the gradients of the largescale fields as we already introduced in the previous chapters. The small aspect ratio in the ocean suggests a separate treatment of lateral and vertical SGS parameterization schemes. We know this closure is based on hypotheses that are only crudely satisfied in the real ocean. Despite the obvious limitations this approach is attractive because the resulting terms are easy to implement and computationally efficient.

8.3.2. Higher Order Closures

More sophisticated strategies are based on the TKE equation derived in previous paragraph. Omitting the molecular viscosity:

$$\frac{\partial k}{\partial t}+\langle u_j\rangle\frac{\partial k}{\partial x_j}+\langle u_i'u_j'\rangle\frac{\partial\langle u_i'\rangle}{\partial x_j}=-\frac{1}{\rho_0}\frac{\partial\langle u_i'p'\rangle}{\partial x_i}-\frac{1}{2}\frac{\partial\langle u_j'u_j'u_i'\rangle}{\partial x_i}-\frac{g}{\rho_0}\langle\rho'u_i'\rangle\delta_{i3}$$

This equation can be integrated in time, provided that the second-order correlations are parameterized without introducing higher order terms. A closed set of equations results by setting (e.g.):

$$\langle u_i'u_j'\rangle=A_M\frac{\partial\langle u_i\rangle}{\partial x_j}$$

$$\langle\rho'u_j'\rangle=A_T\frac{\partial\langle\rho\rangle}{\partial x_j}$$

and parameterizing the triple product term (the dissipation rate of turbulent kinetic energy) as:

$$\epsilon = c_\epsilon \frac{\sqrt{k^3}}{l}$$

Higher-order turbulent closures may also assume that the eddy viscosity and diffusivity coefficients depend on the intensity of the turbulent kinetic energy TKE and a typical length scale in the following combination:

$$A = c_0 l \sqrt{k}$$

Given the TKE equation, the eddy coefficients A_M and A_T, and the coefficients c_ϵ, and c_0, the remaining task is to determine the length scale l. Although the TKE equation can be used for a three-dimensional turbulent closure scheme, it is usually applied to vertical mixing only. Vertical mixing in the ocean is very inhomogeneous and variable: large during convection events, highly variable near the upper and lower boundaries and at steep lateral boundaries, and relatively weak in the interior. So, assuming only vertical component, thus setting all lateral derivatives to zero, the linearized one-dimensional (vertical) version of TKE equation is:

$$\frac{\partial k}{\partial t} + \langle u'w' \rangle \frac{\partial \langle u \rangle}{\partial z} + \langle v'w' \rangle \frac{\partial \langle v \rangle}{\partial z} = -\frac{g}{\rho_0} \langle w'\rho' \rangle - \epsilon$$

Using now:

$$\langle u'w' \rangle = -A_v^M \frac{\partial \langle u \rangle}{\partial z}$$

$$\langle v'w' \rangle = -A_v^M \frac{\partial \langle v \rangle}{\partial z}$$

$$\langle \rho'w' \rangle = -A_v^T \frac{\partial \langle \rho \rangle}{\partial z}$$

we obtain:

$$\frac{\partial k}{\partial t} = A_v^M \left[\left(\frac{\partial \langle u \rangle}{\partial z} \right)^2 + \left(\frac{\partial \langle v \rangle}{\partial z} \right)^2 \right] - A_v^T N^2 + \frac{\partial}{\partial z} \left(A_k \frac{\partial k}{\partial z} \right) - \epsilon$$

where N is the Brunt-Vaisala frequency:

$$N = \frac{g}{\rho_0}\frac{\partial\langle\rho\rangle}{\partial z}$$

and A_k is the vertical diffusivity of the turbulent kinetic energy TKE. The two source terms are related to production by vertical shear and by buoyancy. This equation is the basis for many higher order turbulent closure schemes. The two frequently used approaches are the so-called k-l (Mellor and Yamada, 1982) and k-ϵ schemes. The first uses a length scale to close the system. The second uses a prognostic equation for the dissipation of turbulent kinetic energy. There is no agreement on which closure is preferable for a given situation. Other approaches recognize the fact that the turbulent transports at a given level may not depend exclusively on the local gradients and properties at that level, but rather on the overall state of the boundary layer. A class of parameterizations that takes these processes into account are the non-local "k profile parameterization" (KPP) mixing schemes (Large et al., 1994).

8.4. Lateral Mixing Schemes

Lateral mixing schemes in ocean models usually employ low-order methods. Experience has shown that coarse-resolution models, which do not explicitly resolve the highly energetic mesoscale eddy field, can be run quite successfully with harmonic dissipation terms. For high-resolution simulations, however, where part of the spectrum of mesoscale eddies is explicitly incorporated, the harmonic approach is often found to be too dissipative on the eddy scales, particularly in poorly resolved circumstances in which the cutoff wavenumber on the numerical grid is close to the Rossby deformation radius. The desire to limit the dissipative effects on the eddy field has led to the application of higher-order eddy diffusive and viscous operators.

Since small-scale noise often accumulates at the highest wavenumber, and since these flow components are unlikely to be accurately computed in any case, scale-selective filtering techniques have been developed.

In an inhomogeneous flow field, turbulent viscosity may clearly depend on location (e.g., such variations occur across the Gulf Stream). Since the excluded part of the turbulent spectrum changes with location and local grid spacing, it is plausible that the rate of SGS mixing should vary with location and resolution. A plausible extension of the constant eddy coefficient concept, therefore, is to use spatially varying mixing coefficients. Of course, uniform or simply varying eddy coefficients are crude ad hoc choices. Flow properties and the accompanying levels of turbulent fluctuation are highly inhomogeneous and unlikely to be well represented by simple prescriptions based upon local grid spacing. Another, more advanced class of SGS parameterizations tries to relate these coefficients more directly to ambient conditions in the large-scale flow fields. One straightforward strategy is to keep the form of the operator simple (e.g., harmonic), but to use spatially and temporally varying eddy coefficients which are determined by the local characteristics of the solution. An example of nonlinear lateral diffusion is the scheme proposed by Smagorinsky (1963):

$$A_h^M = \lambda \Delta x \Delta y \sqrt{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2}$$

It combines a grid size dependence with the deformation of the velocity field. The adjustable constant of proportionality in this term is typically set to a value in the range of 0.05 to 0.2.

Thus far, we have implicitly assumed that the turbulent mixing can be thought of as occurring independently in the horizontal and vertical directions. Oceanic mixing, however, is not so easily conceptualized. In the main thermocline, mixing along isopycnals dominates diapycnal mixing. The principle direction of mixing is therefore neither strictly vertical nor purely horizontal, but a spatially variable mixture of the two. In the oceanic interior, it is often expedient to limit excessive diapycnal mixing by orienting the “lateral” mixing along isopycnals.

8.5. Vertical Mixing Schemes

In contrast to lateral mixing strategies, vertical mixing is most often implemented using higher-order closures. The motivation for this derives from the unique importance of the oceanic surface mixed layer, and from the wide variety of vertical mixing processes.

8.5.1. Surface Ekman layer

From a dynamical point of view, the most simple surface boundary layer model assumes a linear viscous momentum coupling to the atmosphere. The solution of this boundary value problem was first obtained by Ekman (1905). It consists of horizontal surface currents which are rotated by 45 degrees to the right (left> relative to the wind direction on the northern (southern) hemisphere at the surface and decay exponentially downward. Note that the Ekman layer is not identical with the mixed layer because it relates solely to the depth of wind influence. Although of similar magnitude, the Ekman depth is not necessarily identical with the mixed layer depth as defined by the vertical variation in density.

8.5.2. Stability dependent mixing

In the vertical, stratification plays a major role in determining the strength of mixing, and a simple extension of the vertically constant mixing coefficient has been widely used. This parameterization scheme, originally suggested by Gargett (1984, 1986), proposes that vertical diffusivity be inversely proportional to the local Brunt-Vaisala frequency.

8.5.3. Richardson number dependent mixing

A vertical diffusion scheme originally developed for the tropical ocean is the Richardson-number-dependent mixing of Pacanowski and Philander (1981). Defining the local Richardson number

$$Ri = \frac{-\frac{g}{\rho_0}\frac{\partial \rho}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2}$$

a general form for the vertical viscous and diffusive coefficient is

$$A_v = \frac{A_{v0}}{(1 + cRi)^n} + A_{vb}$$

where the coefficients c and n are 5 and 2, respectively. A_{v0} is the maximum vertical mixing coefficient, and $A_{vb}$the background viscosity.

Although originally intended for use in tropical oceans, the scheme has been used successfully in mid- and high-latitude oceans as well. In this adaptive and nonlinear parameterization, both the stratification and vertical current shear are considered. For small Ri, the coefficients are large, i e . , strong shear and weak stratification lead to enhanced vertical mixing.

8.5.4. Convection

Static instability of the water column can occur as the result of surface thermohaline forcing especially in high latitudes. In addition, static instability can arise outside the polar oceans by advective processes in upwelling regions along the ocean boundaries, or, on even smaller scales, in fronts. Such statically unstable situations do not persist for a long time; they are removed by vigorous and small-scale vertical motion, called “convective plumes”, which transport the denser water downward, leading to a vertical homogenization of the water column. The compensating upwelling motion does not occur localized but as a large-scale and weaker upward flow. The hydrostatic approximation models have no direct means of producing vertical accelerations in response to occurrences of heavy water over light. Thus static instability has to be removed by other means, since persistent unstable stratification in a numerical primitive equation model would cause internal wave growth, often leading to catastrophic model behavior. Two alternate parameterizations of convection processes are typically used in numerical primitive equation ocean models. One of these, referred to as "convective adjustment" eliminates static instability by an instantaneous vertical mixing of tracers in statically unstable water columns, until all occurrences of unstable stratification are removed.

The other strategy for the removal of static instability involves an increase of the vertical diffusion coefficient in order to diffuse away any unstable stratification by increased vertical mixing. This strategy fits nicely within the framework of both low- and higher-order turbulence closure schemes. Advantages of this approach include improved vectorization, and a temporally and spatially more gradual transition from statically unstable to stable conditions which tends to excite many fewer internal waves than does instantaneous convective adjustment. A possible disadvantage is that static instability may persist over an unphysically long time.

9. EXRCISES

The exercises during this course are done with the NEMO code (https://www.nemo-ocean.eu). The idea is to test different test-cases of the official release, analyse the model results and perform sensitivity tests. Time permitting, we will try to design our own implementation for a specific problem.

9.1. Prepare the environment.

To use the model, we need to download it together with the IO software and install all the needed libraries. To download the IO manager code, called XIOS:

⇒ *svn co http://forge.ipsl.jussieu.fr/ioserver/svn/XIOS/trunk xios*

To download the NEMO code

⇒ *git clone --branch 4.2.0 https://forge.nemo-ocean.eu/nemo/nemo.git nemo_4.2.0*

Now you have both XIOS and NEMO downloaded in your PC (Linux!). Both software to execute properly need a few external libraries, some are standard in most of the Linux distributions others need to be downloaded and installed using specific options. Few details are provided at https://sites.nemo-ocean.io/user-guide/install.html

In the Linux machine available for the course (at the Mat department Levi room) the external libraries have been properly installed in */usr/local/oddo/.* In the DIFA cluster, OPH, the same libraries have been installed in */home/software/terra/NEMO_env/.*

NOTA: in the OPH cluster the XIOS is already installed in */home/software/terra/NEMO_env/xios*. You do not need to download and compile XIOS in the OPH cluster.

9.1. Compile

NEMO and XIOS codes must be compiled for our specific architectures. This process is simplified using configuration files.

Under both NEMO and XIOS main directories you will find a sub-directory named "*arch*". Here you must deploy configuration files specific for your architecture. For XIOS 3 files are needed:

⇒ *arch-DIFA-xios.fcm*
⇒ *arch-DIFA-xios.path*
⇒ *arch-DIFA-xios.env*

For NEMO only 1 file is needed:

⇒ *arch-DIFA.fcm*

If you are using the Levi workstations those files can be downloaded from "virtuale" but need to be modified to match your home directory structure. If you use OPH, XIOS is already compiled, while the configuration file for NEMO named *arch-OPH.fcm* can be found at:

/home/software/terra/NEMO_env/arch_nemo/

Only if you are using the Levi workstation, once downloaded, and copied in the proper directories the following modification is needed:

1) *In arch-DIFA.fcm you need to assign %XIOS_HOME the path where you installed XIOS.*

Now you can compile XIOS and NEMO. XIOS needs to be compiled first and only once.
NEMO must be compiled for every single configuration you want to run.

Only in the Levi workstations:
Move to the directory where you downloaded XIOS
⇒*cd /path/to/xios/*

and type

⇒*./make_xios --arch DIFA-xios*

Now compile NEMO
⇒*cd /path/to/nemo_4.2.0*

in the Levi workstations type:

⇒*./makenemo -m 'DIFA' -r GYRE_PISCES -n 'MY_GYRE'*

In the OPH type:
⇒*./makenemo -m 'OPH' -r GYRE_PISCES -n 'MY_GYRE'*

Executing this command, you copied the reference configuration GYRE_PISCES into a new configuration named MY_GYRE and compiled it. In the OPH cluster.

Summary on OPH:

1. ***Log into OPH***
    1.1. *ssh -J name.lastname@137.204.50.15 name.lastname@137.204.50.71*
2. ***Create a dir for the exercise***
    2.1. *mkdir numlab*
    2.2. *cd numlab*
3. ***Download the NEMO code***
    3.1. *git clone --branch 4.2.0 [https://forge.nemo-ocean.eu/nemo/nemo.git nemo_4.2.0](https://forge.nemo-ocean.eu/nemo/nemo.git)*
4. ***Copy the arch file***
    4.1. *cd nemo_4.2.0/arch*
    4.2. *cp /home/software/terra/NEMO_env/arch_nemo/arch-OPH.fcm .*
    4.3. *cd ..*
5. ***Compile NEMO***
    5.1. *module load ucx/1.13.1*
    5.2. *module load mpi/openmpi/4.1.4*
    5.3. *module load terra/NEMO_env/1.0.1*
    5.4. *./makenemo -m 'OPH' -r GYRE_PISCES -n 'MY_GYRE2'*
6. ***Run the model***
    6.1. *Cd /home/STUDENTI/name.lastname/numlab/nemo_4.2.0/cfgs/MY_GYRE2/EXP00*
    6.2. *Cp /home/software/terra/NEMO_env/ancillary/run_gyre.sh .*
    6.3. *Sbatch run_gyre.sh*
7. ***Run the model with XIOS server***
    7.1. *Cp /home/software/terra/NEMO_env/ancillary/XML/iodef.xml .*
    7.2. *Cp /home/software/terra/NEMO_env/ancillary/XML/file_def_nemo.xml .*
    7.3. *Modify the run_gyre.sh script*
    7.4. *Sbatch run_gyre.sh*
8. ***Download the results (from your PC in a terminal execute)***
    8.1. *scp -o ProxyJump=name.lastname@137.204.50.15 name.lastname@137.204.50.71:numlab/nemo_4.2.0/cfgs/MY_GYRE2/EXP00/tmp_work/GYRE_5d_00010101_00011230_grid_T.nc .*
9. ***To compile all the other experiments copy*** */home/software/terra/NEMO_env/ancillary/compile_nemo.sh*
10. ***Always check on*** */home/software/terra/NEMO_env/ancillary*

9.2. Run the model

Running the model using the HPC facilities provided by DIFA (OPH) requires the setup of a submission script. However, the applications are “light” enough to run also on the Levi workstations.

If you use the Levi workstations, first move to the experiment directory

⇒*cd /path/to/nemo/cfgs/MY_GYRE/EXP00*

and then type:

⇒*mpirun -np 4 ./nemo*

An example of the submission script to be used in OPH cluster is available at:

/home/software/terra/NEMO_env/ancillary/run_nemo.sh

Copy this script in your experiment directory, modify it properly and submit:

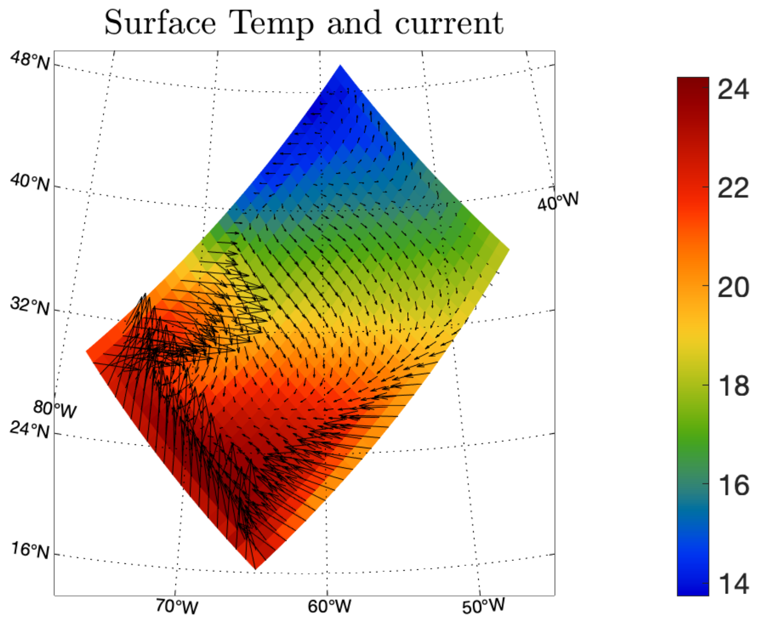
And type:

⇒ *sbatch run_nemo.sh*

9.3. Analyse the results.

The model generated some files in NetCDF format. If we succeeded in using the cluster we need to bring those files locally. After copying the results locally, probably the best is a visual inspection then we can use dedicated software to study them.

9.4. GYRE (using XIOS)


Surface Temp and current
48°N
40°N
32°N
24°N
16°N
80°W
70°W
60°W
50°W
40°W
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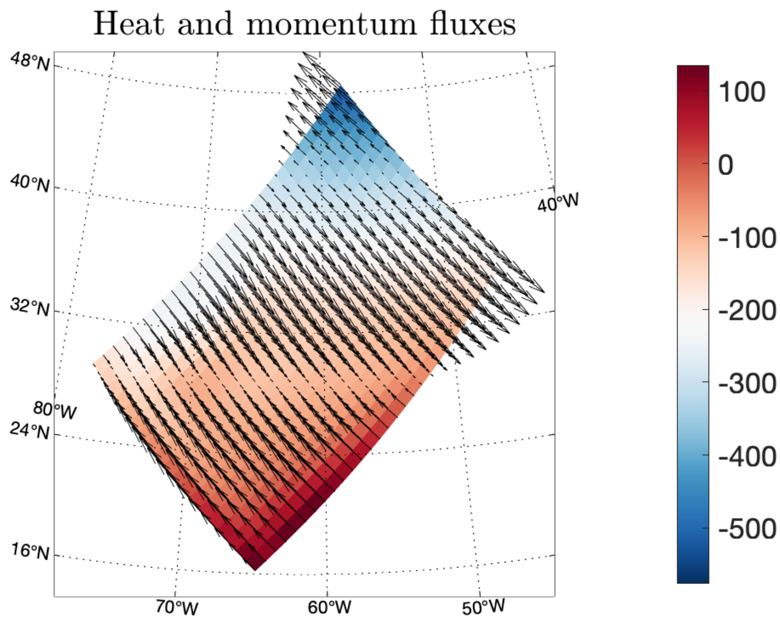
22
20
18
16
14


Heat and momentum fluxes
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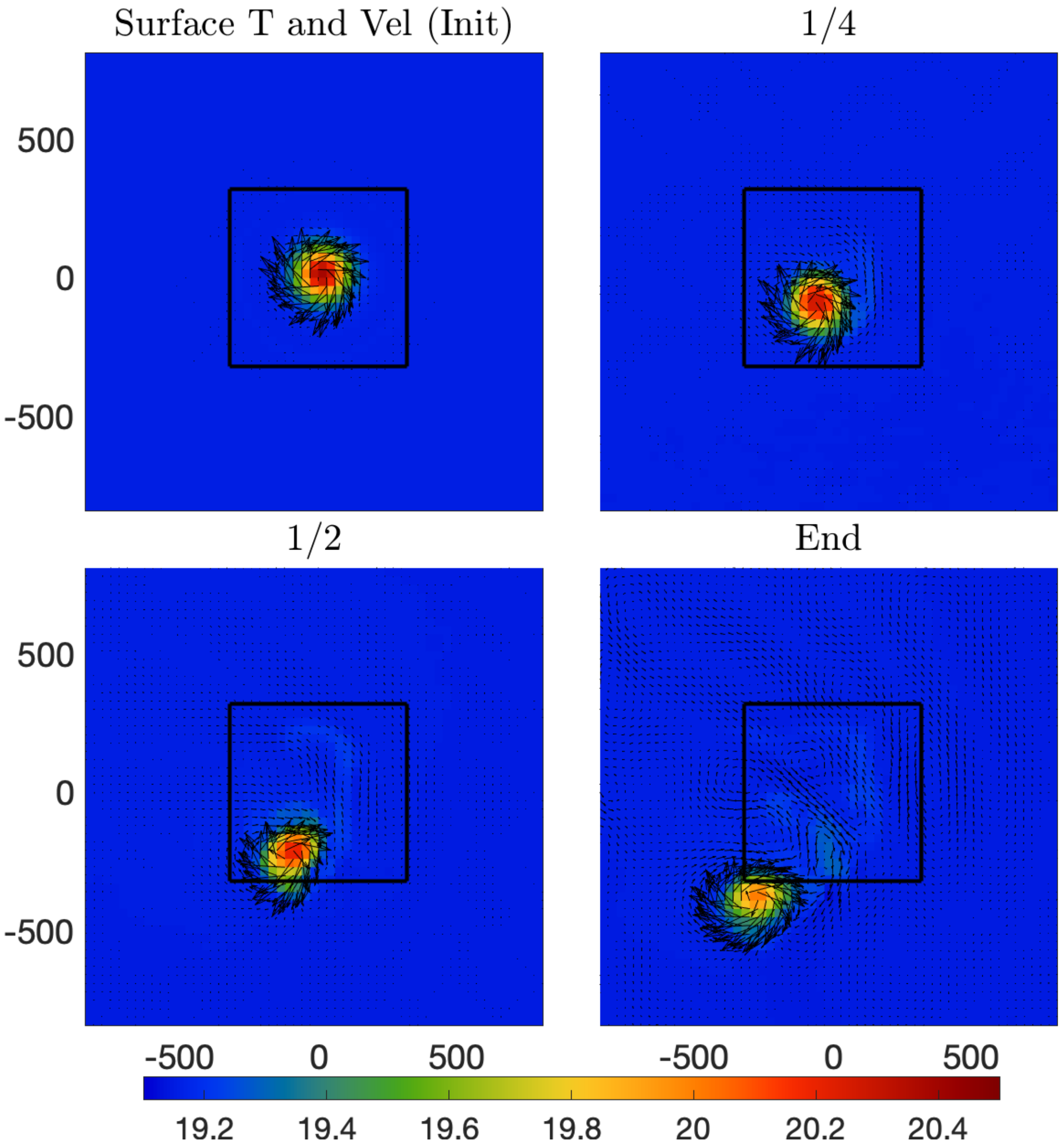
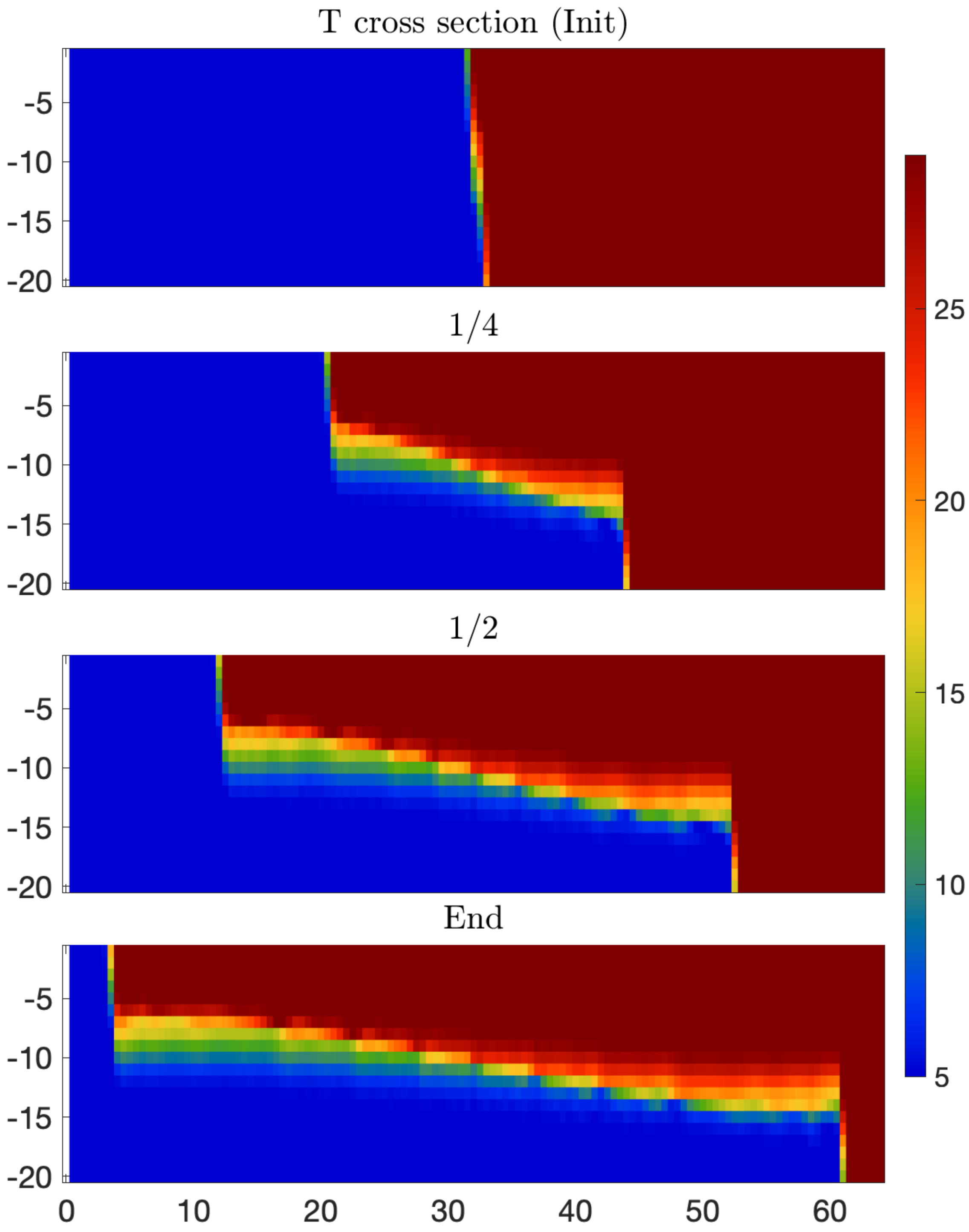
40°N
32°N
24°N
16°N
80°W
70°W
60°W
50°W
40°W
100
0
-100
-200
-300
-400
-500

9.5. VORTEX (two-way nesting)



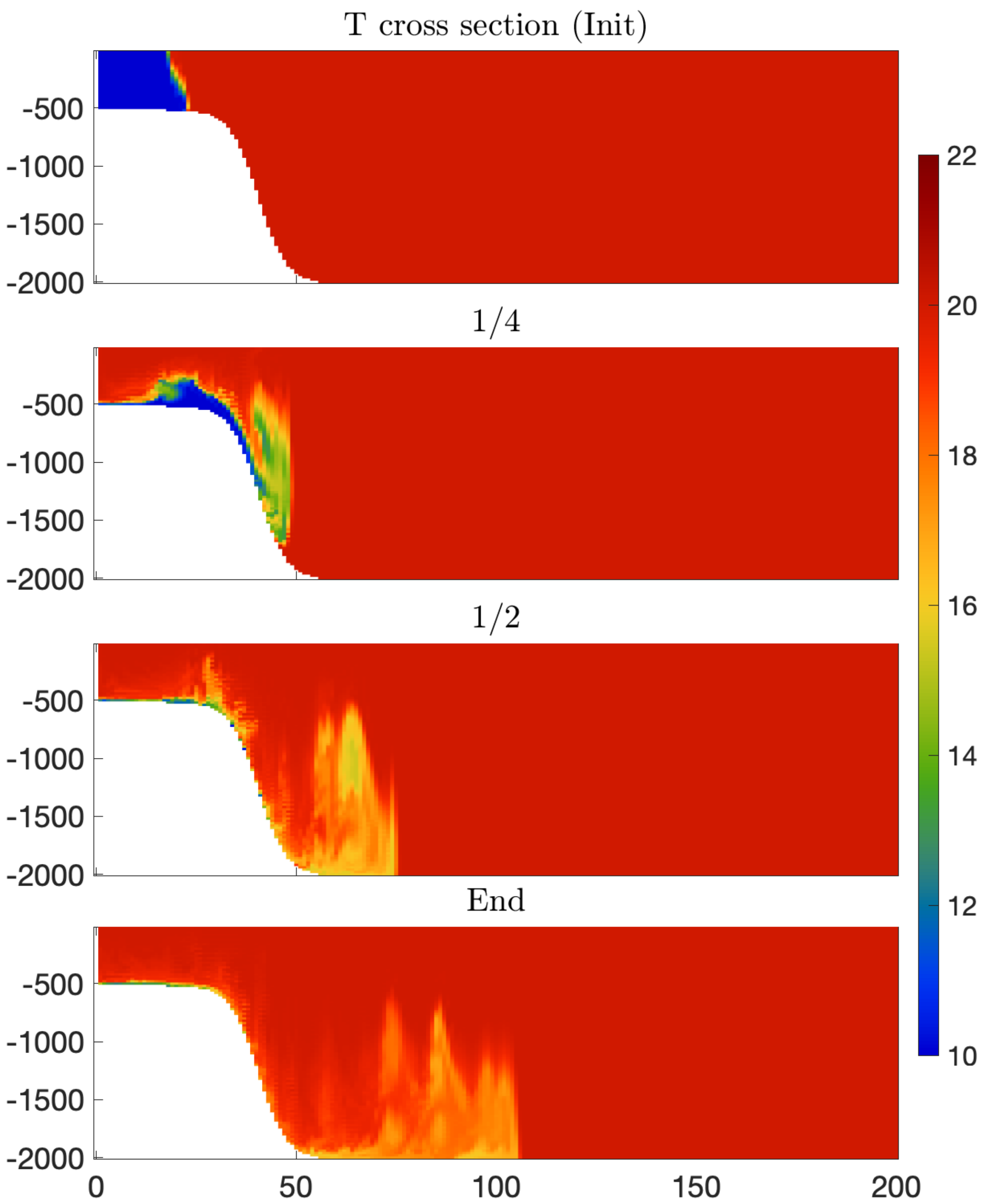
9.6. LOCK_EXCHANGE (advection schemes)

- *D. B. Haidvogel and A. Beckmann. Numerical ocean circulation modeling.* **Series on Environmental Science and Management**, *Apr 1999.* doi:10.1142/p097.
- *M. Ilıcak, A. J. Adcroft, S. M. Griffies, and R. W. Hallberg. Spurious dianeutral mixing and the role of momentum closure.* **Ocean Modelling**, *45-46:37–58, Jan 2012.* doi:10.1016/j.ocemod.2011.10.003.

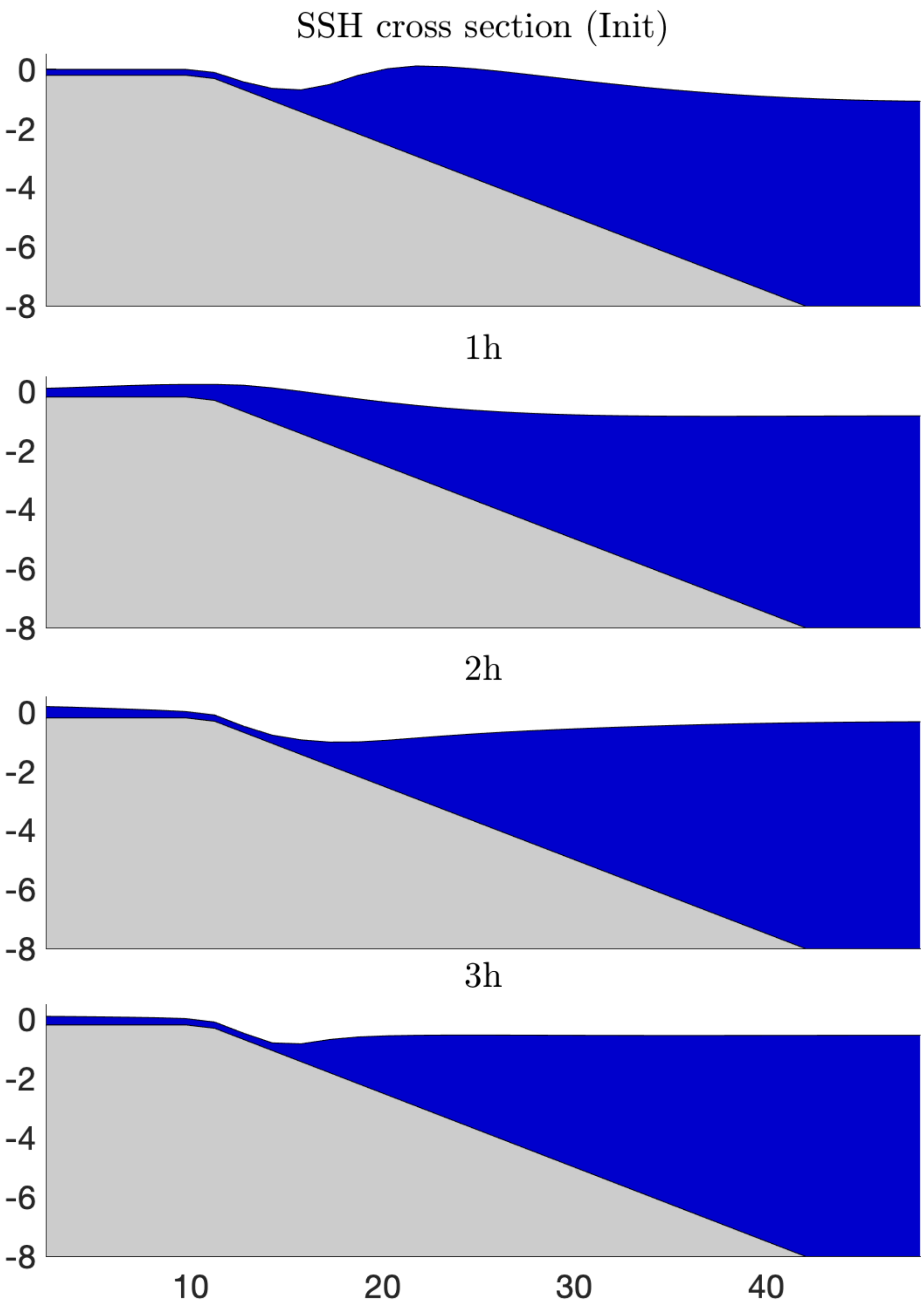


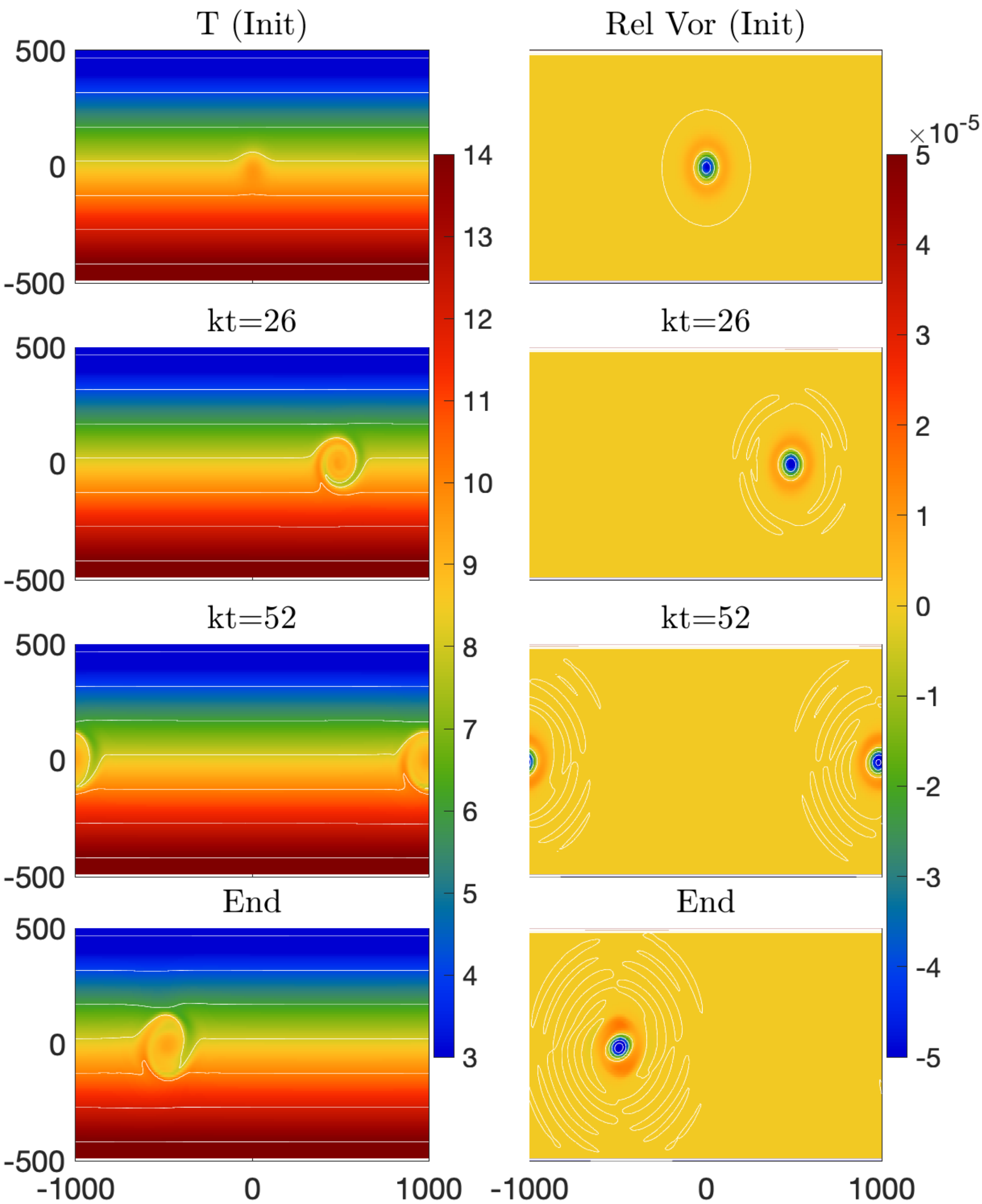
9.7. OVERFLOW

This can be used to test different vertical coordinate system, bottom parameterizations, advection schemes, and diffusion.



9.8. WAD




T (Init)
Rel Vor (Init)
kt=26
kt=26
kt=52
kt=52
End
End
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0
-500
-1000
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11
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8
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1
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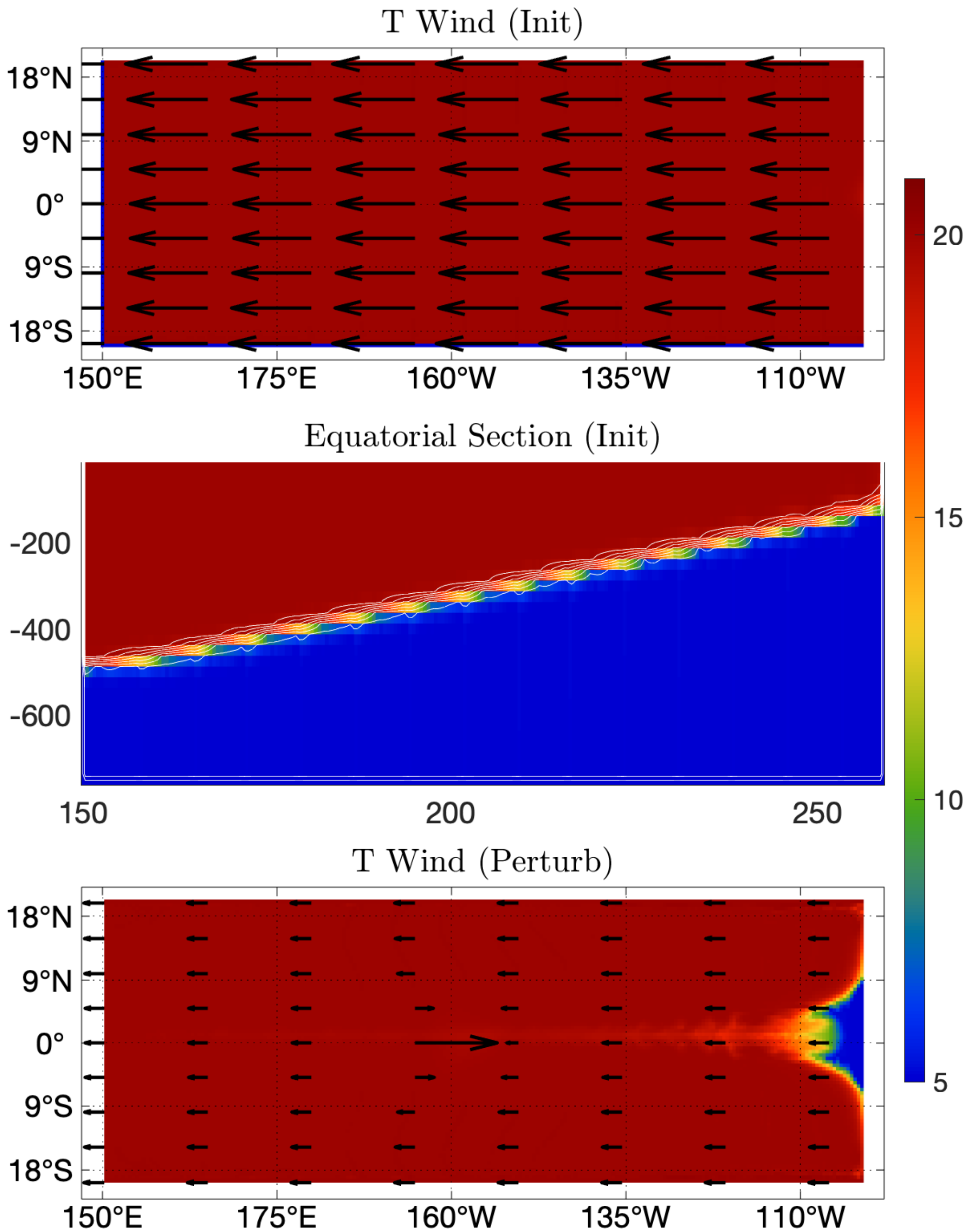
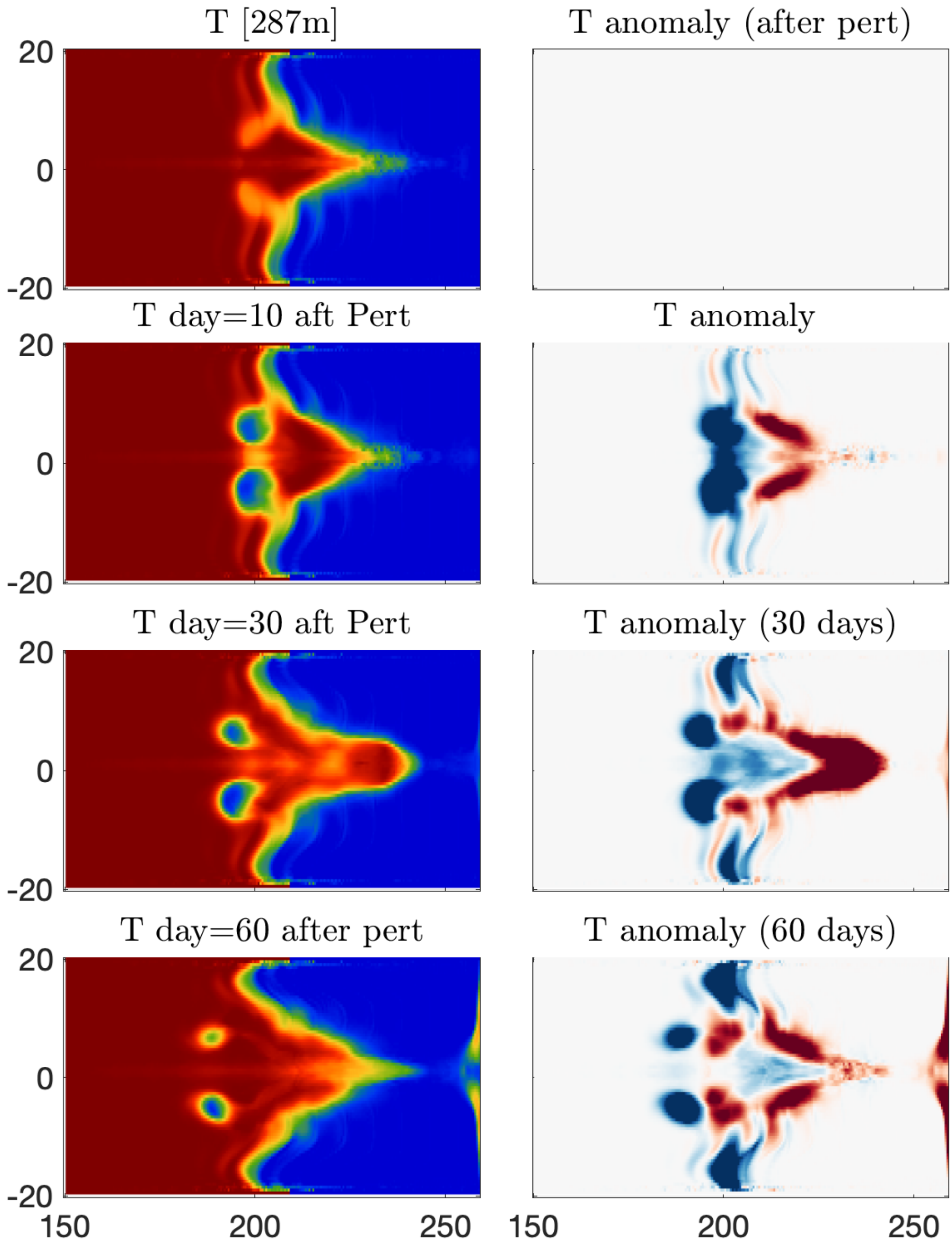
-1
-2
-3
-4
-5

10. EQUATORIAL KELVIN AND ROSSBY WAVES




T [287m]
T anomaly (after pert)
T day=10 aft Pert
T anomaly
T day=30 aft Pert
T anomaly (30 days)
T day=60 after pert
T anomaly (60 days)
20
0
-20
150
200
250

11. APPENDINX

$$e^{\pm i\theta} = cos(\theta) \pm isin(\theta)$$

$$cos(\theta) = \frac{1}{2}\left(e^{+i\theta} + e^{-i\theta}\right)$$

$$sin(\theta) = \frac{1}{2}i\left(e^{+i\theta} - e^{-i\theta}\right)$$

$$sin(2\theta) = 2\,sin(\theta)\,cos(\theta)$$

$$sin^2(2\theta) = 4\,sin^2(\theta)\,cos^2(\theta)$$